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Towards a high resolution Late Palaeoproterozoic – earliest Mesoproterozoic apparent polar wander path for northern Australia

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Seven new primary poles upgrade the apparent polar wander path for northern Australia for the period 1730–1590 Ma. This brings the total number of primary poles to sixteen, of which nine have well-defined SHRIMP U–Pb zircon ages. The upgraded path provides a tool for dating and correlation of tectonic events as well as fluid migration episodes for one of the most important periods of base-metal mineralisation in Australia. Other new primary and overprint poles allow the path to be extended tentatively forward in time to *ca* 1500 Ma and back to *ca* 1770 Ma. The 1730–1590 Ma path comprises five quasilinear segments or tracks, each terminated by a sharp bend. The bends can be related to major (supersequence) boundaries in the northern Australian superbasins and to the initiation of sub-basins that host the base-metal deposits. Most bends are also associated with magnetic overprints related to episodes of fluid circulation, and three appear to be associated with the expulsion of metal-rich brines that formed the world-class Broken Hill, Mt Isa and HYC Pb–Zn deposits. According to the pole path, the movement of the region in the period 1730–1590 Ma can be described as series of clockwise-counterclockwise rotations about mostly similar Euler poles located southwest of Australia, suggesting that the lithosphere may have been coupled alternately to oppositely convecting cells in the mantle. This mode of movement was terminated at the onset of the Isan Orogeny. The oldest track spans a period of igneous activity from 1740 Ma to 1710 Ma that left a 1000 km-long trail of volcanics with a decreasing age trend to the northwest. Palaeomagnetic data are consistent with the interpretation that this trail was caused by the movement of the plate over a thermal anomaly in the upper mantle.

KEY WORDS: apparent polar wander path, McArthur Basin, Mt Isa Inlier, palaeomagnetism, Palaeoproterozoic.

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INTRODUCTION

Palaeomagnetism can provide fundamental information for basin analysis on tectonism, deposition of basin phases, and fluid migration. It can identify some types of alteration event (e.g. distinguish metasomatic from early diagenetic reddening), determine the regional extent of events such as fluid flow, and date phenomena that leave magnetic records. Central to the application of palaeomagnetism is the apparent polar wander path whose shape, in particular, has attracted attention in recent years because of its tectonic implications (Loutit *et al.* 1994; Wyborn *et al.* 1997; Klootwijk 1998; Wyborn *et al.* 1998b). Indeed, the tectonic significance of the shape was recognised as long ago as 1972 by Irving and Park (1972). A sharp bend in the apparent polar wander path signifies an abrupt change in the direction of movement of the plate, implying a change in the system of plate-driving forces and therefore of the stress regime within the plate. The time of tectonism can be determined on an absolute scale or, as is sometimes more important, relative to other geological events that have left magnetic imprints, e.g. magmatism and low-grade metamorphism. Similarly, the timing of sedimentation and of fluid flow can be estimated from the associated remanence directions.

A preliminary Palaeoproterozoic apparent polar wander path for northern Australia was published over a decade ago in a review of Australian Precambrian palaeomagnetism (Idnurm & Giddings 1988). This path, derived from formations of the McArthur Basin, was revised by the inclusion of new poles and by additional measurements to consolidate the earlier poles (Idnurm *et al.* 1995). Nevertheless, gaps remained in the older and younger path segments determined from the lower Tawallah and upper McArthur Groups, respectively. New work, carried out in conjunction with the NABRE program, now extends the sampling to the Lawn Hill Platform and fills those gaps. At the same time, the NABRE sequence-stratigraphic reinterpretation of the basin fill leads to a revision of two earlier primary poles: the pole from the Balbirini Dolomite is split in recognition of a ~10 million year age difference between the upper and lower parts of the formation (Jackson *et al.* 2000), and the pole from the Masterton Formation is reinterpreted to represent a chemical overprint. Especially important for the development of the apparent polar wander path are recent U–Pb SHRIMP ages obtained from zircons in reworked layers of tuffs that were deposited sporadically throughout the genesis of the north-

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ern Australian basins, or from felsic lavas and intrusives (Jackson *et al.* 2000; Page *et al.* 2000). Those dates together with other recent ages from rhyolitic lavas across the region (Page & Sweet 1998) provide a detailed time calibration for the apparent polar wander path. The present paper reports the new results.

GEOLOGICAL BACKGROUND

The McArthur Basin and Lawn Hill Platform are two Late Palaeoproterozoic to Early Mesoproterozoic structural domains of the North Australian Craton (Plumb 1979). Their basements comprise metamorphosed Archaean and Early Palaeoproterozoic complexes (Scott *et al.* 2000) that coalesce to form a ridge-like basement high, the Murphy Inlier, which separates the two domains (Figure 1). To the northwest of the inlier, the McArthur Basin is an aggregate of shelves, sub-basins, fault zones and basement highs (Jackson *et al.* 1987). These include the Batten Fault Zone, which has been the main focus of palaeomagnetic studies. Here, outcrop provides a reasonably continuous section of the preserved Palaeoproterozoic succession. Metamorphism is not evident and deformation is gentle except near faults.

Lying southeast of the Murphy Inlier, the Lawn Hill Platform is a tectonic subdivision of the Mt Isa Inlier. The other subdivisions form a broad mid-Proterozoic multiphase fold belt that ceased to be active at the end of the Isan Orogeny (*ca* 1500 Ma). The Lawn Hill Platform is separated from the eastern terranes by the Mt Gordon Fault, which marks a transition from low- to generally high-grade metamorphism and from mild to strong deformation. NABRE studies indicate that the Lawn Hill Platform is not a single

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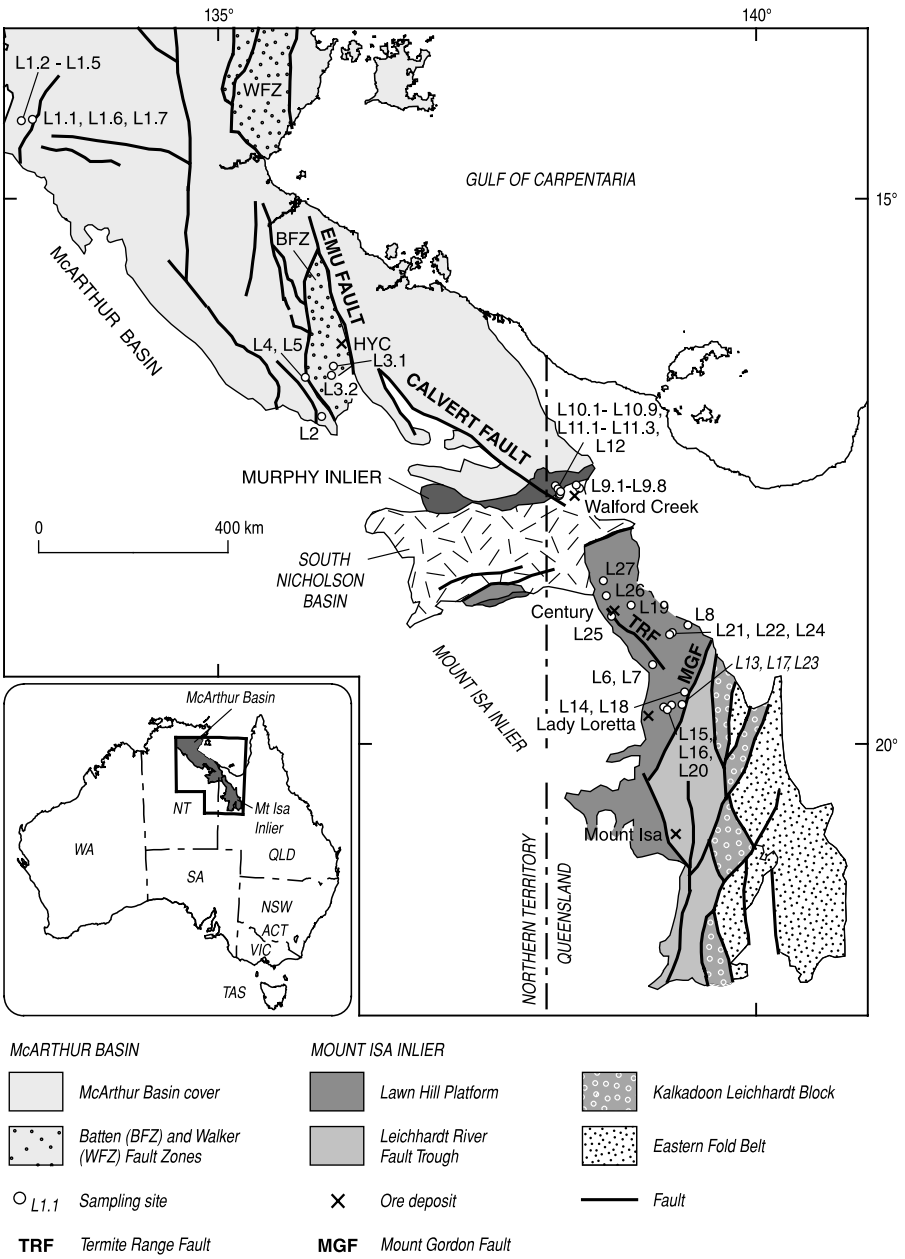


Figure 1 Main tectonostratigraphic subdivisions of central northern Australia showing the localities sampled during the present study. L10.2 refers to site 2 of locality 10, etc.

entity, but rather a complex of superimposed basins that constitute the Leichhardt, Calvert and Isa Superbasins (Jackson *et al.* 2000).

Both the McArthur Basin and Lawn Hill Platform successions comprise siliciclastics, carbonates and bimodal volcanics. Figure 2 shows the respective stratigraphic columns and indicates the formations that were studied. The sequence-stratigraphic framework and the regional tectonostratigraphic history are discussed in the accompanying papers.

FIELD AND LABORATORY PROCEDURES

Unless stated otherwise, the samples were collected according to one of two schemes. Measured stratigraphic sections were sampled systematically at stratigraphic intervals of typically 2 m. Elsewhere samples were collected in groups of four or more from each sedimentary bed, set of closely spaced beds or, in the case of volcanics, each outcrop. The latter were separated by tens of metres to tens of kilometres. The samples were 2.5 cm-diameter cores, obtained with a portable drill, and were oriented usually with both a magnetic and a sun compass. The cores were sandblasted to remove magnetic surface contaminants before being sectioned into 2.2 cm-long specimens.

The measurements were made at the Black Mountain Palaeomagnetic Laboratory in Canberra. Depending on the intensity, the remanences were determined on a 2-axis ScT or 3-axis 2G-Enterprises SQUID cryogenic magnetometer, or a Molspin spinner magnetometer. Thermal

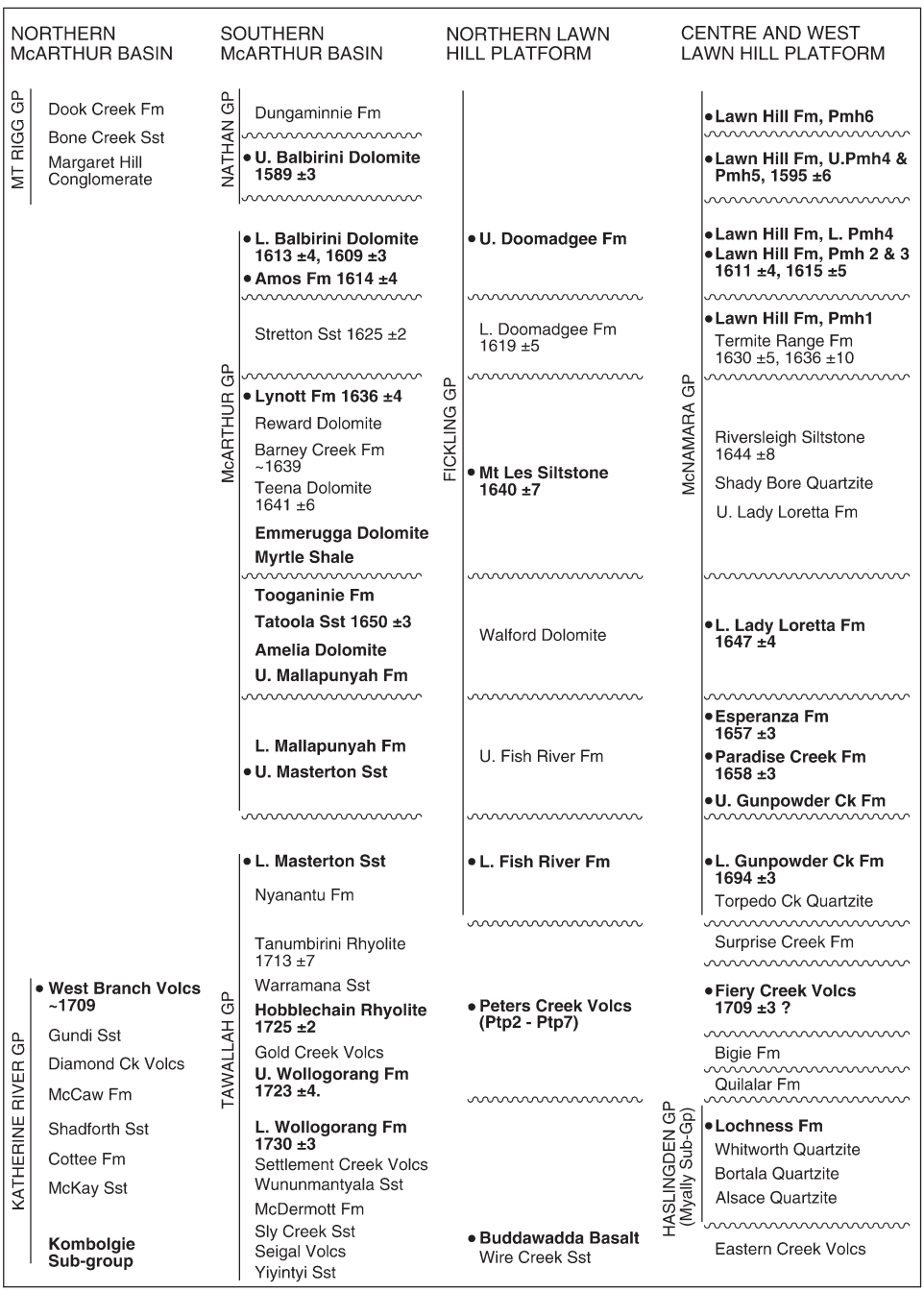


Figure 2 Main Proterozoic stratigraphic units of northern and southern McArthur Basin, and the northern and central-western Lawn Hill Platform. Formations sampled for palaeomagnetism so far are shown in bold; those sampled or updated during the present study are preceded by dots. Regional stratigraphic correlations, shown simplified, are after Rawlings (1999) and Southgate *et al.* (2000). All dates are from U–Pb zircon analyses.

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Table 1 Palaeomagnetic sampling details.

	Locality ID	Lithology	Geogr. coordinates Lat (S)	Long (E)	Map sheet (1:100 000)[a]	Traverse[b]	n[c]	Thickness (m)
West Branch Volcanics								
Locality 1								
Site 1A	L1.1	Rhyolite	14.32	133.26	Waterhouse	–	29	–
Site 1B	L1.1	Arenite	14.32	133.25	Waterhouse	–	17	–
Site 1C	L1.1	Conglomerate	14.32	133.25	Waterhouse	–	6	–
Site 2	L1.2	Rhyolite	14.34	133.13	Waterhouse	–	11	–
Site 3	L1.3	Rhyolite	14.34	133.13	Waterhouse	–	10	–
Site 4	L1.4	Volcanics	14.33	133.13	Waterhouse	–	10	–
Site 5	L1.5	Arenite	14.35	133.16	Waterhouse	–	10	–
Site 6	L1.6	Rhyolite	14.34	133.26	Waterhouse	–	6	–
Site 7	L1.7	Volcanics	14.33	133.25	Waterhouse	–	10	–
Masterton Sandstone								
Locality 2	L2	Quartz arenite	17.10	135.90	Mallapunyah	K78–01	21	15
Lynott Formation								
Locality 3		Dololutites & dolarenites						
Site 1	L3.1		16.65	136.01	Glyde	G77–02	101	150
Site 2	L3.2		16.72	135.97	Mallapunyah	–	12	1
Amos Formation								
Locality 4	L4	Dolomitic siltstones & arenites, tuffs	16.76	137.74	Mallapunyah	M77–03	88	70
Balbirini Dolomite								
Locality 5	L5	Dololutites, dolarenites						
Lower			16.75	135.74	Mallapunyah	M77–04	87	335
Upper			16.75	135.74	Mallapunyah	M77–04	170	562
Lochness Formation								
Locality 6	L6	Sandstones & siltstones	19.36	139.05	Mt Oxide	95JJ–01	20	~5
Fiery Creek Volcanics								
Locality 7	L7	Mafic volcanics & siliciclastics	19.36	139.05	Mt Oxide	95JJ–01	56	~90
Locality 8	L8	Mafic volcanics	18.99	139.48	Mt Oxide	–	18	~2
Peters Creek Volcanics								
Locality 9 (Walford Creek)								
Site 1 (Ptp7)	L9.1	Felsic volcanics	17.77	138.31	Hedleys Creek	–	9	–
Site 2 (Ptp7)	L9.2	Felsic volcanics	17.76	138.31	Hedleys Creek	–	8	–
Site 3 (Ptp7)	L9.3	Felsic volcanics	17.76	138.31	Hedleys Creek	–	9	–
Site 5 (Ptp5)	L9.5	Felsic volcanics	17.73	138.30	Hedleys Creek	–	3	–
Site 6 (Ptp5)	L9.6	Felsic volcanics	17.73	138.30	Hedleys Creek	–	5	–
Site 7 (Ptp5)	L9.7	Felsic volcanics	17.73	138.30	Hedleys Creek	–	10	–
Site 8 (Ptp6s)	L9.8	Sandstone	17.74	138.30	Hedleys Creek	–	4	–
Locality 10 (Wire Creek)								
Site 1 (Ptp5)	L10.1	Felsic volcanics	17.78	138.13	Hedleys Creek	–	16	–
Site 2 (Ptp5)	L10.2	Felsic volcanics	17.77	138.12	Hedleys Creek	–	10	–
Site 3 (Ptp2)	L10.3	Felsic intrusive	17.77	138.11	Hedleys Creek	–	10	–
Site 4 (Ptp2)	L10.4	Felsic intrusive	17.76	138.11	Hedleys Creek	–	20	–
Site 5 (Ptp1)	L10.5	Mafic	17.76	138.11	Hedleys Creek	–	8	–
Site 6 (Ptp1)	L10.6	Mafic	17.75	138.10	Hedleys Creek	–	15	–
Site 7 (Ptp1)	L10.7	Mafic	17.75	138.09	Hedleys Creek	–	15	–
Site 8 (Ptp1)	L10.8	Mafic	17.73	138.09	Hedleys Creek	–	10	–
Site 9 (Ptp1)	L10.9	Mafic	17.74	138.10	Hedleys Creek	–	22	–
Fish River Formation								
Locality 11								
Site 1 (Pff1)	L11.1	Conglomerate (clasts)	17.78	138.13	Hedleys Creek	95AW–01	5	2
Site 2 (Pff1)	L11.2	Sandstone	17.83	138.13	Hedleys Creek	95AW–01	25	33
Site 3 (Pff2)	L11.3	Sandstone	17.79	138.14	Hedleys Creek	95AW–01	10	2
Mt Les Siltstone								
Locality 12	L12	Siltstone & claystone	17.81	138.14	Hedleys Creek	95AW–02	30	25
Gunpowder Creek Formation		Siliciclastics, dolomites & regolith						
Locality 13 (Esperanza W/H)	L13		19.71	139.33	Mammoth Mines	95JJ–02	61	294
Locality 14 (Barr Hole)	L14		19.60	139.36	Mammoth Mines	95PS–01a	78	289
Locality 15 (Paradise Ck 1)	L15		19.75	139.16	Mammoth Mines	96PC–01	11	~20
Locality 16 (Paradise Ck 2)	L16		19.78	139.19	Mammoth Mines	96PC–01a	25	114

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Table 1 continued

	Locality ID	Lithology	Geogr. coordinates Lat (S)	Long (E)	Map sheet (1:100 000)[a]	Traverse[b]	n[c]	Thickness (m)
Paradise Creek Formation		Dolomites, sandstones & shales						
Locality 17 (Esperanza W/H)	L17		19.71	139.34	Mammoth Mines	95JJ–02	22	317
Locality 18 (Barr Hole)	L18		19.60	139.36	Mammoth Mines	95PS–01b	25	23
Locality 19 (Kamarga Dome)	L19		18.82	138.83	Lawn Hill	–	10	~10
Locality 20 (Paradise Ck)	L20		19.73	139.23	Mammoth Mines	96PC–03	30	104
Locality 21 (Mellish Park)	L21		19.09	139.20	Mt Oxide	95PS–02	30	75
Locality 22 (Mellish Park)	L22		19.07	139.23	Mt Oxide	95PS–05	62	300
Esperanza Formation		Siliciclastics						
Locality 23 (Esperanza W/H)	L23		19.71	139.33	Mammoth Mines	95JJ–02	69	183
Locality 24 (Mellish Park)	L24		19.08	139.21	Mt Oxide	95PS–02	21	33
Lawn Hill Formation		Siliciclastics						
Locality 25	L25		18.92	138.63	Lawn Hill	95BJD–09a, b	37	63
Locality 26A (Pmh1, Silver King)	L26		18.74	138.59	Lawn Hill	95BA–07a	81	3.5
Locality 26B (Pmh2, Silver King)	L26		18.75	138.59	Lawn Hill	95BA–07b	8	~10
Locality 27 (Pmh4)	L27		18.60	138.56	Lawn Hill	95BB–09	15	66

[a]Where the samples were collected along a traverse line the coordinates indicate the centre of the locality.
[b]Traverses beginning with letters are in Jackson *et al.* 1987; traverses beginning with numbers refer to NABRE studies.
[c]n is the number of samples.

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demagnetisation was done in large field-cancelled furnaces (McElhinny *et al.* 1971) or in a Schonstedt TSD-1, mu-metal shielded demagnetiser; alternating field (AF) demagnetisation used an automated fixed-axis system incorporated in the 2G-Enterprises magnetometer with in-house modifications (Giddings *et al.* 1997). Magnetic susceptibilities were recorded between heatings to monitor chemical changes in the magnetic minerals. Pilot specimens of some formations were demagnetised by alternating fields, usually in ~20 increments up to 160 mT, as well as thermally to determine which was the more effective of the techniques for separating the remanence components. Most specimens were demagnetised thermally. Depending on the unblocking temperature, 15 to 25 stepwise heatings were required. Specimens from sites that yielded large scatters in remanence directions were first partly demagnetised by AF before thermal treatments in an attempt to remove random components, especially those due to lightning strikes (Schmidt & Embleton 1985). Some samples were pre-treated before demagnetisation by cooling in liquid nitrogen in order to eliminate contributions of low-coercivity, multidomain magnetites to the remanences (Dunlop *et al.* 1997). The directions of the remanence components were obtained from linear segments in orthogonal projection diagrams by principal component analysis (Kirschvink 1980), directions with mean angular deviations (MAD) that exceeded 10° being rejected. Magnetic mineral contents of selected samples were analysed using isothermal remanent magnetisation (IRM) and Lowrie techniques (Lowrie 1990), and by determination of the temperature dependence of low-field magnetic susceptibilities using an AGICO CS-3 Kappabridge.

RESULTS

Four formations were sampled in the McArthur Basin and nine formations in the Lawn Hill Platform, yielding a total of seven primary and 18 overprint poles. The formations, sites and results are described below. Site details, including locality numbers referred to in the text, are summarised in Table 1. The results are listed in Table 2.

McArthur Basin

WEST BRANCH VOLCANICS

The West Branch Volcanics crop out in the northern McArthur Basin (localities L1.1–L1.7, Figure 1). Dated at *ca* 1709 Ma by SHRIMP U–Pb analysis of zircons (Kruse *et al.* 1994), they represent the youngest flows extruded at different centres along a ~1000 km northwesterly band across the Lawn Hill Platform and McArthur Basin during a 20 million year period of mainly felsic magmatism. The thickness of the unit, including sedimentary interbeds, is estimated as 600–850 m (Kruse *et al.* 1994). Like most Proterozoic volcanics in northern Australia, the West Branch Volcanics have been mildly to severely altered.

A total of 109 samples of red-brown volcanic and volcaniclastic samples was collected from seven sites at a ~35 km-long crescent-shaped syncline 80 km east of Katherine. Bedding dips, estimated from sedimentary interbeds at or near the sampling sites, were shallow and lacked sufficient contrast for fold tests. However, cobbles in a conglomerate derived from erosion of volcanics were analysed for a conglomerate test (i.e. to find if their remanence directions are random, as expected for pre-depositional magnetisation).

Normally two or more specimens were demagnetised per sample. Both thermomagnetic analysis (Figure 3a) and remanence intensity behaviour during thermal demagnetisation (Figure 3b) indicated the presence in the volcanics of a mineral with a Curie temperature of ≤580°C,

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Table 2 Summary of palaeomagnetic results.

Formation	NRM int. (mA m^{-1})[a]	n[b]	Comp.[c]	Bedding corr.	Test[d]	Remanence direction Decl.(°)	Inc.(°)	k	α_{95} (°)	Pole Lat.(S)	Long.(E)	K	A_{95} (°)	Acronym	Palaeolat(°)
West Branch Volcanics	5–750	**6**/105/163	A	No		103.7	40.8	37.401	11.1	18.8	202.4	40.249	10.7		23.9
				Yes	C+	*100.4*	*42.9*	*41.205*	*10.6*	*15.9*	*200.5*	*36.042*	*11.3*	WBV	25.3
Masterton Sandstone	1–17	1/23/**23**	A	No	–	*2.4*	*–33.1*	*91.540*	*3.2*	*87.5*	*73.4*	*106.674*	*2.9*	MASO	*28.2*
				Yes	–	357.4	–26.2	89.498	3.2	86.0	277.1	122.550	2.8		24.0
Lynott Formation	0.2–75	**11**/21/23	A	No	–	354.1	–56.7	94.005	4.7	68.6	148.4	49.200	6.6		37.5
				Yes	–	*352.2*	*–48.7*	*82.878*	*5.0*	*75.1*	*162.8*	*54.731*	*6.2*	LYN	29.9
		–/**12**/12	B	No	–	*255.7*	*54.4*	*68.389*	*5.3*	*21.5*	*78.3*	*55.825*	*5.9*	LYNO	35.5
				Yes	–	247.7	47.6	64.238	5.5	27.4	71.0	63.666	5.5		29.4
Amos Formation	0.3–30	1/**39**/41	IT	No	–	350.4	–41.4	36.017	3.9	78.5	184.7	34.428	4.0		24.4
				Yes		*341.3*	*–51.9*	*34.219*	*4.0*	*66.5*	*178.4*	*26.905*	*4.5*	AFB	*33.6*
Balbirini Dolomite, lower	0.1–5	–/**46**/46	IT	No	–	1.0	–35.2	21.955	5.6	86.6	121.8	22.061	4.6		20.1
				Yes		*341.1*	*–51.8*	*21.782*	*4.6*	*66.1*	*177.5*	*14.503*	*5.7*	BDBL	*33.6*
Balbirini Dolomite, upper	0.1–5	–/**44**/47	IT	No	–	6.4	–43.0	17.252	5.3	78.4	106.1	14.786	5.8		26.8
				Yes	–	*326.5*	*–60.2*	*17.056*	*5.4*	*52.0*	*176.1*	*9.319*	*7.5*	BDBU	*42.6*
Peters Creek Volcanics	5–520	**9**/69/69	A	No		289.0	–27.6	75.986	5.9	22.3	219.0	133.171	4.5		14.9
Upper (Ptp5–Ptp7, felsic)				Yes	C+, R+, R_a	*293.0*	*–26.4*	*88.344*	*5.5*	*26.0*	*221.0*	*114.986*	*4.8*	PCVU	*14.9*
		5/**35**/35	B	No	–	*265.6*	*72.7*	*97.338*	*2.5*	*17.5*	*105.3*	*33.612*	*4.2*	PCVO	*58.6*
				Yes	–	239.9	72.4	97.216	2.5	30.9	106.1	33.656	4.2		58.1
Peters Creek Volcanics	5–270	1/**8**/8	C	No	–	99.1	8.7	35.249	9.5	10.0	226.8	117.071	5.1		4.4
Lower (Ptp1, dolerite?)				Yes	–	*100.0*	*5.4*	*35.204*	*9.5*	*10.3*	*228.8*	*119.475*	*5.1*	PCVI	*2.7*
Peters Creek Volcanics	5–400	2/**5**/5	D	No	–	308.1	–8.6	89.740	8.1	37.5	236.7	125.177	6.9		4.3
Lower (Ptp1, basalt?)				Yes	–	*308.6*	*–2.1*	*90.022*	*8.1*	*36.8*	*240.8*	*128.130*	*6.8*	PCVB	*1.0*
Fish River Fm (Pff1)	10–160	1/18/**18**	A	No	–	*262.1*	*70.5*	*38.756*	*5.6*	*18.9*	*101.5*	*14.854*	*9.3*	FSHO2	*55.2*
				Yes	–	227.8	67.9	38.702	5.6	38.62	101.5	16.912	8.7		52.0
Fish River Fm (Pff2)	1–10	1/8/**8**	B	No	–	*179.9*	*38.6*	*76.661*	*6.4*	*85.8*	*141.7*	*78.420*	*6.3*	FSHO3	*22.0*
				Yes	–	178.0	26.0	76.876	6.4	85.5	290.7	97.061	5.7		13.8
		1/9/**9**	**C**	No	–	133.6	63.1	67.390	6.3	43.0	181.9	36.476	8.6		45.4
				Yes	–	*143.0*	*51.5*	*67.446*	*6.3*	*53.9*	*196.8*	*52.527*	*7.2*	FSHO1	*32.6*
Mt Les Siltstone	0.5–11	1/21/**24**	A	No	–	*1.6*	*–37.7*	*157.671*	*2.4*	*86.3*	*114.1*	*152.100*	*2.4*	MLSO	*21.2*
				Yes	–	0.2	–24.9	157.529	2.4	85.3	320.6	194.358	2.1		13.1
Lochness Formation	10–100	1/12/**20**	A	No	–	*6.8*	*–40.6*	*39.558*	*5.3*	*82.3*	*86.8*	*38.511*	*5.3*	LNSO1	*24.0*
				Yes	–	21.5	–14.7	39.606	5.3	66.0	22.8	60.791	4.2		7.6
		1/12/**14**	B	No	–	111.7	2.0	54.789	5.4	20.7	235.6	75.937	4.6		1.0
				Yes	–	*118.9*	*24.2*	*54.789*	*5.4*	*31.2*	*225.6*	*63.151*	*5.0*	LNSO2	*12.7*
Fiery Creek Volcanics	20–450	3/7/**8**	A	No	–	*1.0*	*–38.7*	*284.316*	*3.3*	*87.3*	*119.6*	*280.634*	*3.3*	FCVO	*29.6*
				Yes	–	17.4	–18.5	284.170	3.3	70.5	21.1	434.908	2.7		9.5
		5/37/44	B	No	–	97.8	12.0	68.244	9.3	9.4	225.9	72.510	9.1		6.1
				Yes	–	*108.0*	*40.1*	*67.567*	*9.4*	*23.9*	*211.8*	*55.029*	*10.4*	FCV	*22.9*

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Table 2 continued

Formation	NRM int. (mA m^{-1})[a]	n[b]	Comp.[c]	Bedding corr.	Test[d]	Remanence direction Decl.(°)	Inc.(°)	k	α_{95} (°)	Pole Lat.(S)	Long.(E)	K	A_{95} (°)	Acronym	Palaeolat(°)
Gunpowder Creek Fm	1–100	3/37/**37**	A	No	F–	*250.0*	*66.8*	*30.031*	*4.4*	*27.4*	*97.1*	*14.078*	*6.5*	GPRO1	*50.7*
				Yes		122.3	51.9	8.485	8.6	34.1	195.6	6.024	10.5		38.5
		2/11/**11**	B	No		171.7	68.6	3.589	28.2	56.3	150.9	2.094	43.0		52.3
				Yes	F+	*123.4*	*54.2*	*27.588*	*8.9*	*38.5*	*198.9*	*17.125*	*11.4*	GPRO2	*35.7*
		3/37/**37**	C	No	F–	*1.9*	*–39.2*	*71.667*	*2.8*	*86.7*	*105.4*	*63.513*	*3.0*	GPRO3	*22.4*
				Yes		338.0	–22.4	7.785	9.0	69.3	240.9	14.391	6.4		14.3
		1/19/**19**	D	No	–	287.1	–35.3	52.561	4.7	22.1	215.5	43.066	5.2		19.6
				Yes	–	*282.1*	*–1.1*	*52.479*	*4.7*	*11.6*	*232.9*	*68.544*	*4.1*	GPRR	*0.6*
Paradise Creek Formation	0.5–15	1/8/**9**	A	No	–	*359.0*	*–39.6*	*145.943*	*4.3*	*87.0*	*157.4*	*156.783*	*4.1*	PDCO1	*22.6*
				Yes	–	4.0	–40.6	146.143	4.3	84.8	94.5	155.243	4.2		23.4
		–/**20**/24	B	No	F–	*197.4*	*55.8*	*30.279*	*6.0*	*66.9*	*103.1*	*21.407*	*7.2*	PDCO2	*37.5*
				Yes		166.7	64.9	5.551	15.3	59.5	147.6	2.971	23.1		49.8
Esperanza Formation	0.5–10	–/12/**12**	A	No	–	*358.7*	*–42.7*	*193.949*	*3.1*	*84.6*	*152.7*	*203.719*	*3.1*	ESPO	*25.0*
				Yes	–	315.8	–13.9	10.900	13.8	45.1	239.1	12.931	12.6		7.2
Lawn Hill Formation	0.1–30	–/10/**10**	A	No	–	*234.7*	*66.9*	*100.612*	*4.8*	*37.1*	*97.3*	*40.424*	*7.7*	LWNO1	*49.7*
				Yes	–	180.5	66.1	34.303	8.4	58.3	136.9	16.245	12.4		50.6
		–/23/**25**	B	No	F–	*5.1*	*–38.6*	*118.669*	*2.7*	*84.4*	*80.5*	*127.942*	*2.6*	LWNO2	*22.0*
				Yes		8.4	–33.8	18.317	7.0	81.7	48.7	17.487	7.1		19.0

Italics indicate the alternative chosen because of field test results or as preferred interpretation. k, K are precision parameters (Fisher 1953); α_{95}, A_{95} are semi-angles of 95% cone of confidence.

[a]Range, excluding outliers.

[b]No. of sites/samples/specimens; bold indicates the number used for mean remanence direction and pole calculation.

[c]Remanence component referred to in text.

[d]F, C, R denote fold, conglomerate and magnetostratigraphic reversal test; + and – indicate if the test was positive or negative. R_a denotes positive reversal test of McFadden & McElhinny (1990) at classification A.

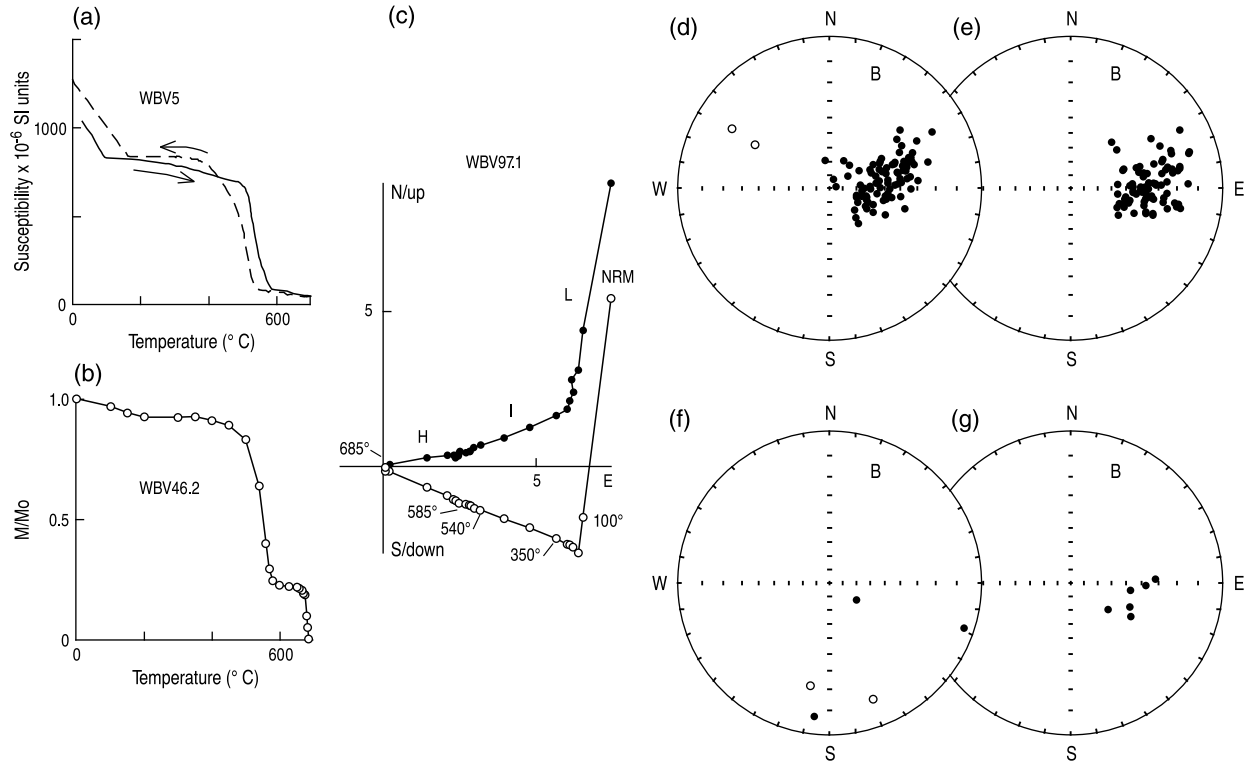


Figure 3 West Branch Volcanics. Example of sharp temperature changes of susceptibility (a) and normalised remanence intensity (b) of the volcanics at ~580°C, indicating the presence of minerals with unblocking temperatures ≤580°C and >580°C. Orthogonal projection plot (c) reveals the presence of three components—L (low temperature, recent field), I (<580°C) and H (>580°C)—in the remanence. Note small break in slope between I and H at ~580°C. Equal-angle stereoplots (d) and (e) show, respectively, the I and H directions. The I directions form a distinctly elongated group due to incomplete separation of components. Stereoplot (f) shows the large scatter of remanence directions between five West Branch Volcanics cobbles in a conglomerate deposited between flows, and plot (g) shows the site mean directions for the H component from which the primary pole was calculated for the formation. In this and the following stereoplots the open (filled) symbols indicate directions in the upper (lower) hemisphere, and B signifies that the directions have been bedding corrected. In the orthogonal projection plots open (filled) symbols indicate projection onto vertical (horizontal) plane. Units along the axes are mA m^{-1}.

probably magnetite, and another that unblocked at 600–680°C, probably hematite. Many of the orthogonal projection plots showed small breaks in slope at ~580°C (Figure 3c). Therefore, in the analysis, the remanences above and below ~580°C were treated as different, even though for some specimens this temperature did not signal a distinct change of direction. Figure 3d, e shows the ≤580°C and >580°C directions, excluding those that deviated by >30° from the group means. The distribution of directions with ≤580°C unblocking temperatures is streaked, indicating the presence of two components. Comparison with Figure 3e suggests that one of these may be the higher temperature (600–680°C) component, while the other is an overprint that affected magnetite but not the higher Curie temperature hematite. Since thermal demagnetisation failed to adequately separate the components, the ≤580°C directions are discarded.

Figure 3f shows the remanence directions in five volcanic cobbles from the interflow conglomerate mentioned above. Laboratory treatments and analysis for the cobbles were the same as for the outcropping volcanics. Although highly scattered, the directions are strongly biased towards the southeast quadrant, suggesting partial remagnetisation of what were originally random directions. The randomness indicates a positive conglomerate test. A likely cause of the partial remagnetisation is heating by younger flows deposited on top of the conglomerate. Since the remagnetisation would have occurred during the period of emplacement of the West Branch Volcanics package, its remanence is taken to be primary.

The >580°C unblocking temperature directions group reasonably well. Their site means are shown in Figure 3g. Some of the scatter in this figure may be attributed to errors in bedding dips, which were often difficult to estimate accurately. The overall bedding-corrected mean direction for the West Branch Volcanics, calculated from the site means, is declination 100.4°, inclination 42.9°.

MASTERTON SANDSTONE

The previously sampled Masterton Sandstone in the southern McArthur Basin (Idnurm *et al.* 1995) was revisited to sample in detail the upper 15 m of the formation at the original locality. This interval contains fine, red-brown mottles that coalesce locally to form highly ferruginous zones. The earlier results indicated that the ferruginisation is associated with a magnetic overprint (MASO) which is similar to, but not identical with, the *ca* 1640 Ma overprint OP2 in the southeastern McArthur Basin. The difference between the Masterton Sandstone overprint and OP2 mean

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directions appeared sufficiently large to tentatively regard the Masterton Sandstone overprint as separate, OP4, even though the result was based on only 14 sample directions, and OP2 was found in other, nearby successions.

Twenty-one new samples were collected to consolidate the previous results and to confirm the separate identity of OP4. As previously, thermal demagnetisation yielded mostly a single remanence component. The new directions are indistinguishable from the earlier directions (Figure 4), and their combined mean, declination 2.4°, inclination –33.1° ($\alpha_{95} = 3.2°$), seems sufficiently well defined to establish OP4 as distinct from OP2. As discussed by Idnurm *et al.* (1995), the ferruginisation was probably caused by fluids migrating in a deep aquifer formed by the Masterton Sandstone and sealed by the overlying fine-grained, dolomitic McArthur Group. From the position of the pole on the apparent polar wander path the age of the ferruginisation appears to be *ca* 1500 Ma.

Because its directions differ widely from those of the top ferruginous zone, the remanence of the lower Masterton Sandstone was previously believed to be primary. However, the formation has recently been found to contain a major unconformity with a counterpart in the Lawn Hill Platform where it divides the Gunpowder Creek Formation into upper and lower packages (Southgate *et al.* 2000). A regolith developed on the Gunpowder Creek unconformity locally contains a ferruginous phase, which, as discussed below, gives remanence directions similar to those of the lower, red-brown Masterton Sandstone. The remanence of the Masterton Sandstone is therefore reinterpreted as an early overprint, formed during the break in deposition, and the palaeomagnetic pole MAS in Idnurm *et al.* (1995) is renamed MASO2.

LYNOTT FORMATION

The Lynott Formation of the upper McArthur Group comprises in decreasing age the Caranbirini, Hot Spring and Donnegan Members. Sampling focused on the stromatolitic Hot Spring Member, which comprises dololutites and fine- to medium-grained dolarenites. Reworked tuffs in this member have been dated by U–Pb zircon analysis at 1636 ± 4 Ma (Page *et al.* 2000). Localities sampled for palaeomagnetism were the Lynott Formation type section in the southern McArthur Basin (Glyde 77/02 *in* Jackson *et al.* 1987) and a site 7 km southeast of the type section. At the type section the outcrop is discontinuous, with resistant beds forming rock platforms in gullies and benches on hill slopes. In fresh exposures the colours range from pure grey through light-brown to red. Bedding dips are shallow. The sites spanned a distance of 1 km in the direction of dip, corresponding to a stratigraphic interval of ~150 m. The second locality, near a fault, was selected because its moderately steep bedding presented the possibility of a fold test. The outcrop, a single ~1 m-thick resistant dolarenite redbed, was sampled along strike.

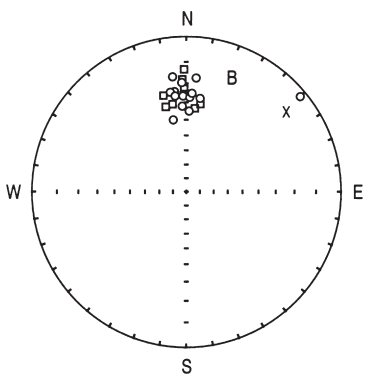


Figure 4 Masterton Sandstone. Overprint remanence directions in ferruginised zone below Mallapunyah Formation boundary. Outlier x was excluded from the pole calculation. Square symbols represent directions obtained during the present study; circle symbols represent directions obtained previously (Idnurm *et al.* 1995). Symbols as in Figure 3.

A total of 113 samples was collected. The red samples were found to contain single-component remanences that unblocked at typically ~680°C and had westerly downward inclinations (Figure 5a). The directions plot as an elongate group for both *in situ* and bedding-corrected orientations (Figure 5c, d). For both orientations the McFadden fold test statistic ξ_2 (McFadden 1990) is 16.1. This value exceeds the 99% confidence limit of 7.8 for a positive fold test, indicating that the directions are statistically different. One explanation for the difference is that the successions at the two localities acquired their magnetisation in different periods. Alternatively, the results may be explained by further tilting after magnetisation at the second locality because of movement at the adjacent fault zone. Because of this uncertainty, the results from the second locality are rejected. The directions from the first locality resemble closely those of a previously identified overprint, OP3, which appears to be widespread in the southern McArthur Basin (Idnurm *et al.* 1995).

Only a few of the grey and light-brown samples had stable remanences. These contained mostly a single component that unblocked in the temperature range 400–480°C. Their inclinations were mostly northerly, moderately steep, and upward (Figure 5b, e). These directions are consistent with the Late Palaeoproterozoic apparent polar wander path and yield a pole 75°S, 163°E that plots on the apparent polar wander path in stratigraphic sequence with other McArthur Basin poles. A recent viscous origin for the magnetisation seems to be ruled out by the 400°C+ unblocking temperature as well as by a significant departure from the present field direction (Figure 5e). This component is therefore interpreted as primary.

AMOS FORMATION

The Amos Formation was sampled at its type section where an ~80 m-thick succession of red and grey siltstones and sandstones grades upwards to massive dolostones, which are capped by karst. Pink, redeposited tuffaceous beds in the middle of the formation contain zircons dated by U–Pb at 1614 ± 4 Ma (Page *et al.* 2000).

A total of 88 samples was collected from benches of resistant outcrop. Preference in sampling was placed on red clastics and pink tuffs in the middle ~10 m of the formation. An earlier, reconnaissance, study indicated that grey samples become magnetically unstable during demagnetisation. Applied each on its own, AF and thermal techniques failed to separate the remanence components in the red

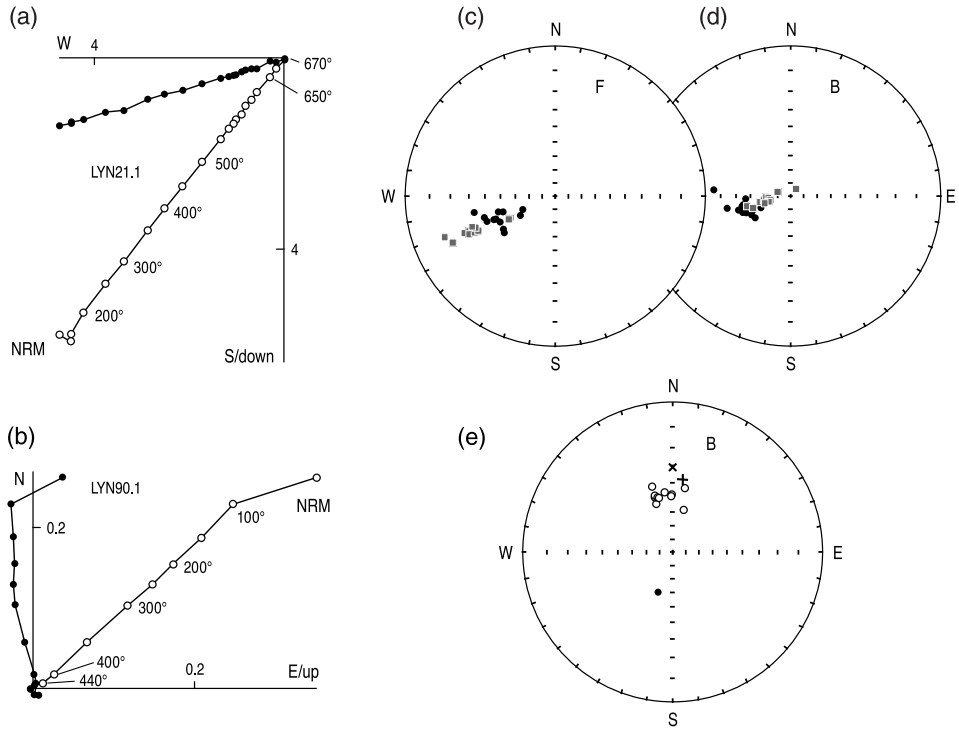


Figure 5 Lynott Formation. (a) and (b) are typical orthogonal projection plots for red and grey dolomitic samples. Plots (c) and (d) show remanence directions from the red samples before (F) and after (B) bedding corrections, both groups being distinctly elongate. The directions from the site adjacent to a fault are indicated by squares. Plot (e) shows site mean remanence directions for grey and light-brown samples. Heavy cross and plus indicate the present dipole and present field directions at the sampling locality. Symbols as in Figure 3.

siliciclastics. However, a combination of AF followed by thermal demagnetisation proved effective (Figure 6a, b), revealing three components. The first (LC) had a low coercivity, high unblocking temperature and random directions—a characteristic of remanences due to lightning strikes which may be expected at the exposed sampling site. The second component (LT) was removed at the initial (100°C) heating step, suggesting that it is carried by goethite and is therefore probably modern. The third, often small, component (A) unblocked in the temperature range 400–650°C and had a moderate upward inclination to the north (Figure 6c). Its tilt-corrected mean direction is declination 341.3°, inclination –51.9°. Although a fold test is not available, this component is interpreted as primary because the pole plots in stratigraphic sequence relative to other poles on the apparent polar wander path. The Amos pole lies close to the pole from the lower Balbirini Dolomite (*ca* 1611 Ma), consistent with the similarity of ages of the two formations.

BALBIRINI DOLOMITE

A primary palaeomagnetic pole has been reported previously for the Balbirini Dolomite (Idnurm *et al.* 1995). However, the NABRE sequence-stratigraphy work reveals a major depositional break in the middle of the formation (Jackson *et al.* 2000). This is confirmed by the SHRIMP U–Pb dates of 1609 ± 3 Ma and 1613 ± 4 Ma for zircons from

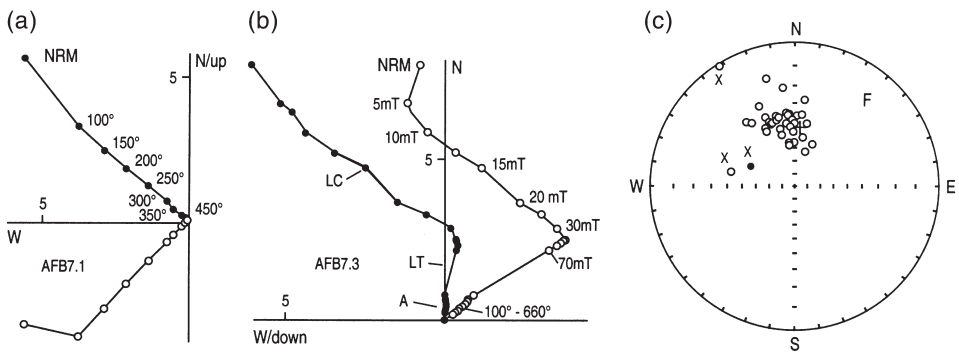


Figure 6 Amos Formation. Orthogonal projection plots show thermal (a) and alternating field (AF)-thermal (b) demagnetisations of two separate specimens from the same sample. Thermal treatment yielded only a single, spurious component, whereas AF followed by thermal treatment yielded three components: LC, low coercivity (probably lightning induced); LT, low temperature (present-day goethite remanence); and A, interpreted as primary remanence. Stereodiagram (c) shows the A component directions. Cross indicates present geomagnetic field direction and x indicates outlier directions omitted from the pole calculation.

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two redeposited tuffaceous beds in the lower part of the formation and 1589 ± 3 Ma for zircons from a single tuff bed in the upper part (Jackson *et al.* 2000). Because of the large age difference, a single primary pole is no longer appropriate for the formation. The original data have therefore been reprocessed, yielding a bedding-corrected mean remanence direction with declination 341.1°, inclination –51.8° for the lower Balbirini Dolomite and declination 326.5°, inclination –60.2° for the upper Balbirini Dolomite (Figure 7).

Whereas the Balbirini Dolomite previously yielded a positive McFadden (1990) fold test, splitting the formation into two parts leaves too few directions for conclusive fold tests. Nevertheless, it should be noted that the upper directions fail only marginally to return a positive test at the 95% confidence level. In favour of the primary origin of the remanences, the two sets of directions differ significantly, with Watson's (1956) test indicating a <1% probability that the directions are from the same population. A stratigraphic thickness of ~900 m was covered by the sampling, which should ensure that secular variations of the geomagnetic field have been averaged out and are therefore not the cause of the directional difference. This difference would rule out the presence of a ubiquitous regional overprint. The poles plot in stratigraphic order on the north-trending apparent polar wander path.

Lawn Hill Platform: North

PETERS CREEK VOLCANICS

The bimodal Peters Creek Volcanics, which include sediments and intrusives, are subdivided into seven members Ptp1 to Ptp7 (Sweet *et al.* 1981). The basal Ptp1, named Buddawadda Basalt, contains mafic intrusives. It is separated by a regional unconformity from the overlying felsic suite which is likely to be much younger (Scott *et al.* 1998). Ptp2, Ptp5 and Ptp7 are mostly intrusive complexes (Jackson *et al.* 2000). U–Pb SHRIMP zircon dates are available for the lower (1729 ± 2 Ma) and upper (1726 ± 2 Ma) parts of Ptp2 (Jackson *et al.* 2000), and for Ptp7 (1725 ± 2 Ma) (Page & Sweet 1998).

The formation was sampled at 17 sites within a broad belt of scattered outcrop on the southern flank of the Murphy Inlier where the volcanics are ~2 km thick. The sites were along the upper and middle Wire and Walford Creeks, ~20 km apart. Also sampled were subangular felsic clasts of the Peters Creek Volcanics in the basal conglomerate of the Fish River Formation. At Wire Creek the latter overlies Ptp5 with a small angular unconformity. Except for Ptp3 and Ptp4, which were not sampled, the lower members of the Peters Creek Volcanics were sampled at Wire Creek and the upper members at Walford Creek, Ptp5 being sampled at both localities. Samples from the Ptp1 intrusive were grey-brown; all other samples including those from arenite interbeds were pale to medium reddish brown. Bedding dips (~10°) were estimated from nearby sedimentary rocks. Demagnetisation was thermal. Some specimens from localities that gave large directional scatters were first partially AF demagnetised in an attempt to eliminate the scatter should that be caused by lightning strikes.

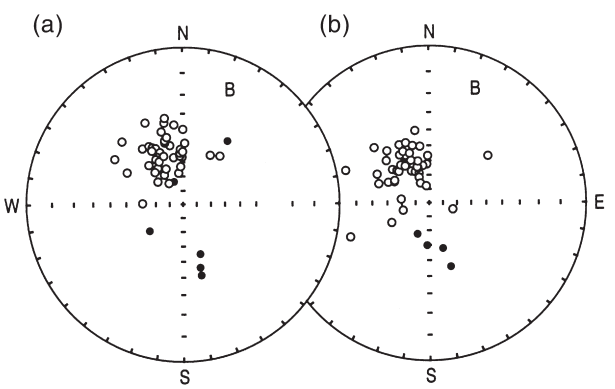


Figure 7 Balbirini Dolomite. Remanence directions from the lower (a) and upper (b) dolomite. Symbols as in Figure 3.

Ptp2 yielded scattered remanence directions, and the arenites yielded directions that appear to be broadly consistent with overprints OP2, OP4 or late Tertiary weathering. Preferential remagnetisation of the arenites may be attributed to their high porosity relative to that of the igneous rocks. The remanence of felsic samples from the other members unblocked generally at 685–700°C, indicating hematite as remanence carrier. For most felsics the orthogonal projection plots were linear from ~300°C to 690°C; however, for some the intensity–temperature plots showed pronounced inflections at ~580°C, suggesting the presence of a second, magnetite, component (Figure 8a). Usually these inflections corresponded to small slope changes in the orthogonal projection diagrams, but the net difference was too small to be statistically significant. This suggests that the magnetite and hematite were magnetised at, or approximately, the same time. The felsic remanences (component A, pole PCVU in Table 2) formed bipolar directional groups in the east-southeast and west-northwest quadrants of the stereoplot (Figure 8b), and passed the McFadden and McElhinny (1990) reversal test at classification A. The bedding-corrected mean, after inversion of the west-northwest directions, was declination 110.7°, inclination 24.8°. A more conservative mean, calculated from the nine site mean directions, is declination 113.0°, inclination 26.4°. Two lines of evidence support a primary origin for this remanence. First, the rhyodacite clasts in the conglomerate deposited on top of the rhyodacitic Ptp5 have random remanence directions (Figure 8c), suggesting that this lithology has not been overprinted. The clasts were subjected to the same laboratory treatments and analysis as the samples from Ptp5. Second, a geomagnetic reversal occurs between Ptp5 and Ptp6, again implying the absence of blanket magnetic overprinting. The polarities in Ptp5 are the same at the Wire and Walford Creeks sites and are opposite to those in the other members.

Scarcity of unweathered outcrop, pervasive chemical alteration, and difficulty in the absence of field relationships of distinguishing the basalt flows from shallow dolerite intrusives posed problems in the sampling of Ptp1. The alteration is characterised by hematisation with variable degrees of destruction of magnetite. Thermal demagnetisation yielded three remanence components: B, C and D. Component B with steep west downward dipping directions and an unblocking temperature of ~685°C was found at all sites (Figure 8d). Because a similar remanence

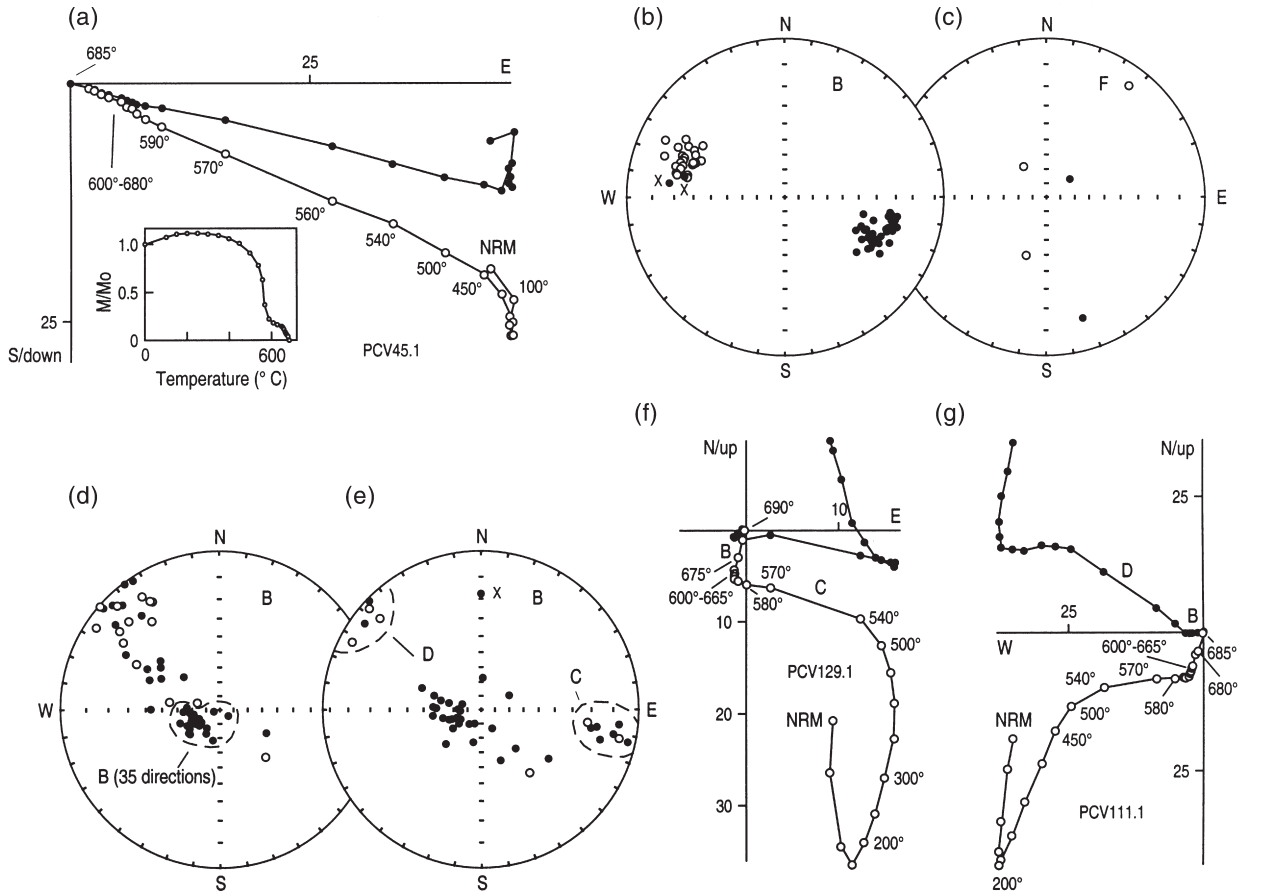


Figure 8 Peters Creek Volcanics. Orthogonal projection plot (a) shows a small change at ~580°C in a typical felsic sample after removal of magnetite remanence component. Inset in (a) shows the corresponding decrease in remanence intensity at ~580°C, confirming the presence of magnetite in the red felsic members of the Peters Creek Volcanics. Tight clustering of remanence directions in the felsic samples in (b) contrasts the randomness of the directions in felsic clasts within a conglomerate that lies immediately on top of the Peters Creek Volcanics (c). The latter samples provide a positive conglomerate test. Stereoplot (d) shows the high-temperature B component in the Buddawadda Basalt and a northwest band of pseudo-directions in a dolerite that intrudes the Buddawadda Basalt. Plot (e) shows magnetite temperature range directions in the mafics, including components C and D discussed in the text. Plots (f) and (g) are examples of orthogonal projection plots for mafics that contain components B, C and D. Outliers x in the stereoplots were excluded from pole calculations. Symbols as in Figure 3.

direction is encountered in the overlying Fish River Formation as well as in other formations of the Lawn Hill Platform, B appears to be an overprint. Component C with shallow downward inclinations to the east (Figure 8e) was found at only one site (L10.7) where it was the dominant remanence above 550°C (Figure 8f). C unblocks with a sharp decrease in intensity at 570–585°C, indicating magnetite as the carrier.

Component D, which unblocked at ≥570°C, was found at two sites (L10.6 & L10.9). Its intensity–temperature plots, like those of the felsic Peters Creek Volcanics, contained pronounced inflections at ~580°C. These two demagnetisation characteristics suggest that D resides in both magnetite and hematite, and therefore the data for <580°C and >580°C were analysed separately. The former data yielded a cluster of five subhorizontal northwest directions that define D (Figure 8e, g). The latter yielded a band of directions between B and D groups (Figure 8d), indicating an incomplete separation of those components. To avoid bias the >580°C directions were excluded from the pole calculation.

The presence of component D in both magnetite and hematite provides a clue to the origin of the remanence components in Ptp1. It suggests that the magnetisation is a thermoremanence, acquired probably when the lava cooled below the Curie point of magnetite (magnetite remanence), and that this remanence was later superimposed by a chemical component acquired during deuteric alteration (hematite remanence). Component D is therefore interpreted as the primary remanence of the Buddawadda Basalt. The difference in direction between C and D suggest that they were acquired at different times, and C must therefore be younger than D. C is found only in magnetite, consistent with an intrusion that was not exposed to a deuteric alteration environment during emplacement. It is therefore tentatively interpreted as the primary remanence of the dolerite.

FISH RIVER FORMATION

The Fish River Formation at the base of the Fickling Group, northwestern Lawn Hill Platform, contains three members:

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Pff1 (predominantly sandstone), Pff2 (predominantly shale) and Pff3 (predominantly sandstone). A major unconformity separates Pff2 from Pff3.

The formation was sampled at Wire Creek where it lies unconformably on member Ptp5 of the Peters Creek Volcanics. Samples were collected from two stratigraphic intervals. The first was the basal 35 m of Pff1, which starts with the 2 m-thick conglomerate mentioned in the previous section. Otherwise this succession comprises mainly red-brown, cross-bedded fluvial sandstones. The second interval was a dark red-brown, weathered (ocherous), medium-grained sandstone ~10 m below the Pff2/Pff3 unconformity. Because of the predominance of hematite the samples were demagnetised thermally.

The difference in the stratigraphic settings of the two sampling sites is reflected in their remanences. The Pff1 site contains component A (pole FSHO2 in Table 2) that unblocks in the range 560–670°C and has a steep west-downward inclination (Figure 9a). Component A resembles the overprint B in the Peters Creek Volcanics and is therefore likewise an overprint.

The Pff2 site contains component B (pole FSHO3 in Table 2) that unblocks at 640–660°C and is bipolar with a moderate north-upward/south-downward inclination (Figure 9b). B is based on few directions and has a large confidence circle making it difficult to distinguish from Pliocene or younger chemical weathering effects, and from a common Mesoproterozoic overprint (OP4) in the Lawn Hill Platform. Another component, C, at this site unblocks at ~680°C and has a steep downward inclination to the southeast (Figure 9c). As discussed later, the C directions yield a pole FSHO1 that plots near the *ca* 1650 Ma bend on the apparent polar wander path. Because Pff2 correlates with the 1694 ± 3 Ma lower Gunpowder Creek Formation (Southgate *et al.* 2000) C is inferred to be an overprint. The results indirectly indicate the presence also of component A at this site. The evidence is in the form of elongation of a group of steep, downward-dipping directions, which are spread between the A and C directions (Figure 9d). This suggests that the remanence is a mixture of the A and C components, therefore implying the presence of overprint A at the Pff2 site. Complete separation of A and C components in the remanence is made difficult by the similarity of their unblocking temperatures and directions.

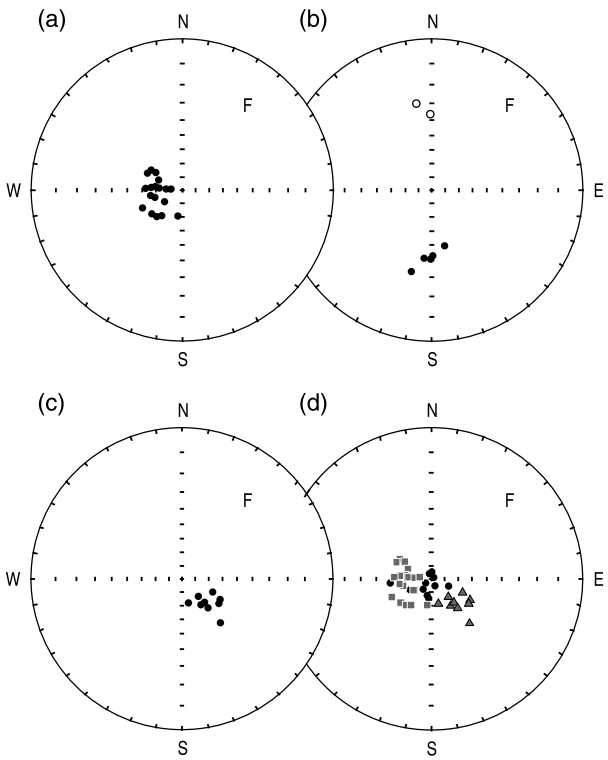


Figure 9 Fish River Formation. Remanence directions for components A (a), B (b) and C (c). Filled circles in (d) indicate directions interpreted as pseudo-directions, since they are intermediate between component A (squares) and C (triangles) directions. A and C appear to have similar unblocking temperature spectra. Symbols as in Figure 3.

MT LES SILTSTONE

The Mt Les Siltstone comprises siliciclastics and subordinate carbonates. Also present are tuffaceous beds containing zircons that have been dated by the U–Pb SHRIMP technique at 1640 ± 7 Ma (Page & Sweet 1998). The formation was sampled in the lower ~25 m of a cliff where Wire Creek has exposed pale greenish-grey and cream siltstones and claystones. Demagnetisation was thermal.

Initial remanence intensities of the samples were low (typically 1–3 mA/m), and were further reduced by removal at the initial 100°C heating step of a random component. Most of the remaining remanences unblocked at 540–630°C and had moderate north-upward/south-downward inclinations (Figure 10a). Three samples contained remanences of both polarities. The mean direction before tilt correction (declination 1.6°, inclination –37.7°) is similar to that of the Mesoproterozoic overprint OP4 found throughout the Lawn Hill Platform, as well as Pliocene directions related, for example, to chemical weathering. The latter possibility is unlikely because the component is present in seemingly unweathered green-grey samples. Consequently, the overprint is interpreted as OP4.

Six samples from the base of the cliff contained evidence of a moderately steep, west-downward-directed component (Figure 10b), suggesting the presence of an OP3-like

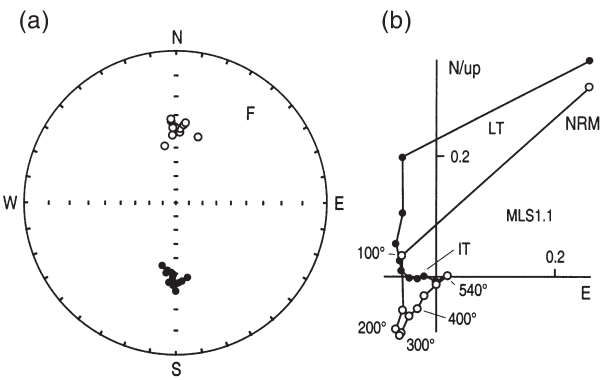


Figure 10 Mt Les Siltstone. Stereoplot (a) shows bipolar remanence directions characteristic of the sampled section and orthogonal projection plot (b) shows a spurious low unblocking temperature component LT together with west downward-inclined component IT found in some samples at the base of the section. Symbols as in Figure 3.

overprint. As noted above, the latter was found also in the nearby Peters Creek Volcanics and Fish River Formation.

Lawn Hill Platform: Centre and West

LOCHNESS FORMATION

The Lochness Formation, pre-dating the Calvert Superbasin (Jackson *et al.* 2000) is the oldest unit sampled in the McArthur Basin and Lawn Hill Platform. Its age is poorly constrained by the date 1790 ± 9 Ma from the stratigraphically older Bottletree Formation (Page 1983) and 1709 ± 3 Ma from the younger Fiery Creek Volcanics (Scott *et al.* 1998). At its Seymour River sampling locality the Lochness Formation comprises red siltstones and fine-grained sandstones. The beds dip 37° to the southeast.

Twenty samples were collected from the bed and side cliffs of the Seymour River. Thermal demagnetisation yielded two remanence components, A and B. Component A unblocked at 650–675°C and its *in situ* remanence had a moderate upward inclination to the north (Figure 11a, c). This direction recurs throughout the overlying McNamara Formation, and is therefore most likely an overprint. In most samples its removal revealed a shallow east component B (Figure 11c, d). The red colour of the rock and truncation of the Lochness Formation by an erosional unconformity that separates the McNamara Group from the older strata suggest that B may be a chemical, possibly weathering-related, overprint acquired during the depositional hiatus. Such an origin would be consistent with the interpretation, discussed below, that the Buddawadda Basalt pre-dates the unconformity. Alternatively, B could be an earlier overprint or even the primary remanence. In view of the large tilt of the beds, it is important to know if bedding corrections should be applied to the remanence directions. As discussed later, component A post-dates the main stages of the Isan Orogeny; therefore tilt corrections have not been applied. On the other hand, component B being older than 1730 Ma, is likely to pre-date most of the deformation, and has therefore been tilt corrected.

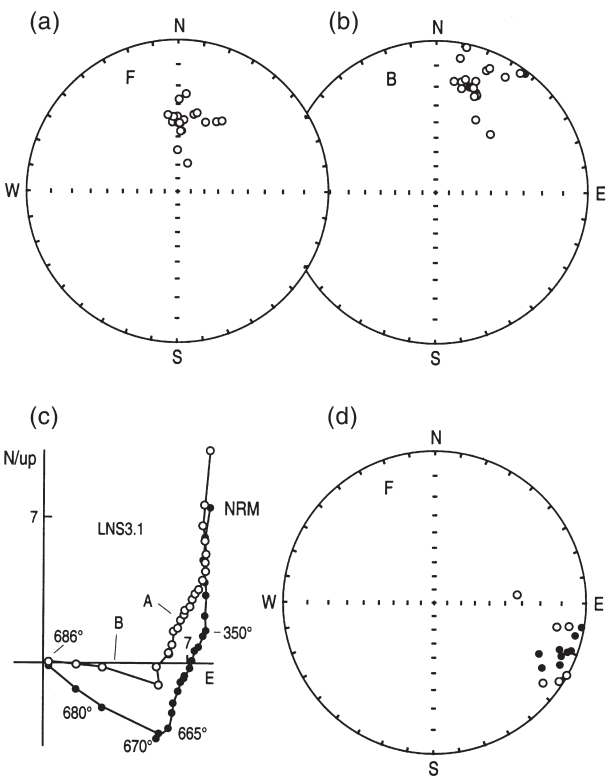


Figure 11 Lochness Formation. Field (a) and bedding (b) corrected directions for component A. Field corrected (*in situ*) directions correspond to a pervasive overprint at the Lawn Hill Platform. Bedding corrections yield a pole that does not plot on any known apparent polar wander path segment and these corrections are therefore considered inapplicable. Orthogonal plot (c) shows components A and B, B being resolved only in the last 15°C interval of demagnetisation. Stereoplot (d) shows B component directions. Symbols as in Figure 3.

FIERY CREEK VOLCANICS

The bimodal Fiery Creek Volcanics were sampled mostly at the same locality as the Lochness Formation where only the lower parts of the volcanics, comprising mafic flows and interbedded siliciclastics, are found. Both rock types were sampled. Also sampled were small remnants of a mafic flow at a second locality ~60 km to the northeast (L8, Figure 1). The bedding dip at the second site (~ 40° to 320°), although very poorly defined, differs greatly from the dip at the first locality (35° to 155°), providing a rough fold test. Except for a SHRIMP zircon date of 1709 ± 3 Ma for what appears to be a late felsic intrusive (Scott *et al.* 1998), the Fiery Creek Volcanics lack chronological constraints.

A total of 74 red- and grey-brown samples was collected, 56 of these from the first locality. Thermal demagnetisation yielded two components, A and B, that unblocked generally between 680°C and 690°C and formed coherent directional groups. Component A was found only in the interbedded sediments, and its directions resemble closely those of component A of the Lochness Formation. As noted above, the Lochness component appears to be an overprint.

Component B (Figure 12a, b) was found in both the siliciclastics and volcanics. Before tilt corrections its directions for the two localities differed widely (Figure 12a). Although tilt corrections did not produce coincidence, the directional difference was greatly reduced (Figure 12b). While the remaining discrepancy may be attributed to an age difference between the sampled units or to continued deformation after magnetisation, it is more likely to be due to an error in the bedding dip at the second locality. The fold test is therefore considered to be positive. Because of the uncertain bedding dip estimate at the second locality only the directions from the first locality were used in the final analysis. In calculating the overall mean direction for the formation, unit weights were assigned to individual flows and interbedded sediment packages, giving the tilt-corrected direction: declination 108.8°, inclination 40.1°. This resembles closely the mean direction obtained by assigning unit weights to each of the samples: 109.7°, 39.8°. Some volcanic samples contained evidence of a possible third, magnetite-resident, component that unblocked at 580°C and had directions similar to those of B (Figure 12a).

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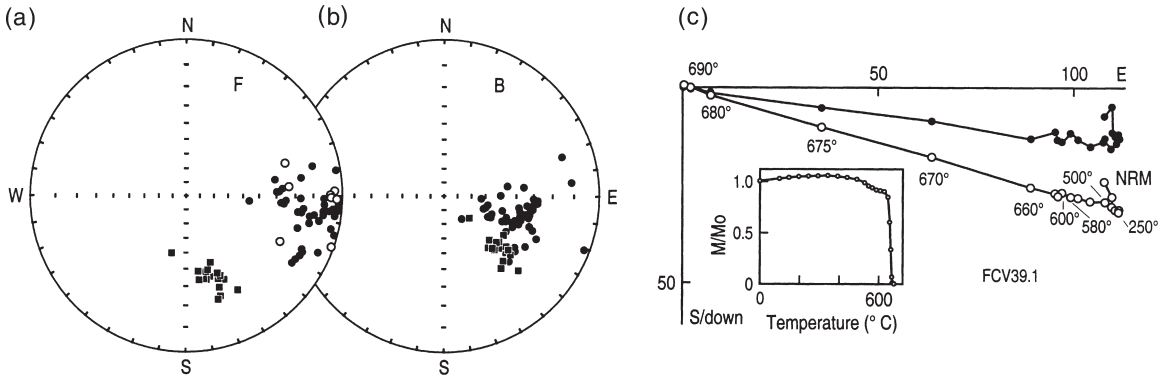


Figure 12 Fiery Creek Volcanics. Stereodiagrams show field (a) and bedding (b) corrected directions for component B at two localities distinguished in the diagrams by circle and square symbols. Bedding corrections improve the directional agreement between the localities, but retain some differences. The latter are attributed to errors in the bedding dip estimate for the site represented by the square symbols. Orthogonal projection plot (c) shows a small change in the remanence direction at ~600°C. Inflection in the normalised remanence intensity plot (inset) at this temperature suggests that the change is due to the removal of a small magnetite remanence. Symbols as in Figure 3.

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That component was, however, too poorly defined to give a reliable pole.

GUNPOWDER CREEK FORMATION

The Gunpowder Creek Formation near the base of the McNamara Group consists of fine-grained, shallow-marine siliciclastics and minor dolomites. The formation is divided into two sedimentary packages by an erosional surface that represents a depositional break of 20–25 million years (Southgate *et al.* 2000). At one locality this surface is covered by a 3 m-thick ferricrete; at others it contains conglomerates. Zircons from a tuffaceous bed in the lower Gunpowder Creek Formation give a SHRIMP U–Pb age of 1694 ± 3 Ma (Page *et al.* 2000). The age of the upper Gunpowder Creek Formation is constrained by U–Pb zircon dates ranging from 1658 ± 3 Ma to 1653 ± 7 Ma in the conformably overlying Paradise Creek Formation (Page *et al.* 2000). Resumption of sedimentation after the long depositional break is attributed to thermal relaxation following emplacement of the 1678 ± 2 Ma Carters Bore Rhyolite and 1671 ± 8 Ma Sybella Granite (Southgate *et al.* 2000).

Samples were collected along measured stratigraphic sections traversing the formation. Two sections (Esperanza Waterhole and Barr Hole) were near the Mt Gordon Fault zone; the third, near Paradise Creek, was ~10 km west of the fault (L15 & L16, Figure 1). Moderately steep to near-vertical bedding dips at the first two localities contrast with moderate dips at the third and provide a good opportunity for fold tests. Rock colour ranged from dark grey to buff and, less commonly, red-brown. The relative effectiveness of thermal and AF demagnetisations was tested on pilot specimens. For some specimens AF demagnetisation was applied after thermal (100°C) or liquid nitrogen pretreatment. Thermal demagnetisation was found to give less scatter in orthogonal projection plots and was therefore adopted for routine work.

Four remanence components, A–D, were resolved in the Gunpowder Creek Formation. Component A was found at all three sampling localities, where it occurred in both grey and buff samples. It unblocked at 450–580°C and formed a west-downward inclined group in a background of highly scattered directions (Figure 13a). Tilt corrections dispersed the group (Figure 13b), indicating that the magnetisation is post-deformational, i.e. an overprint. Its mean direction (declination 250.0°, inclination 66.8°) gives the pole GPRO1. The latter falls near the OP3 group of overprints.

Component B unblocked at ~685°C and was found in 11 sandstone samples from the upper Gunpowder Creek Formation. Although collected at two widely separated localities (L13 at Esperanza Waterhole and L15 in the Paradise Creek region) the samples were from the same stratigraphic unit (they straddle the Gun 2.1f surface of Southgate *et al.* 2000). At both sites the beds were distinguished from drab adjacent lithologies by their dark red-brown colour. Before tilt corrections the directions from these localities differed widely; after corrections they converged (Figure 13c, d), indicating that this component predates the Isan and earlier orogenies. The B component pole (GPRO2) lies on the apparent polar wander path at the younger side of the lower Mallapunyah Formation pole MAL, suggesting that B could be an overprint. This possibility is reinforced by the dark red-brown colour of the beds, which suggests that B is probably due to hematite introduced from migrating fluids. Moreover, gamma-ray counts from this stratigraphic interval are unusually low, consistent with alteration due to circulation of K-poor fluids. As noted in the discussion, the circulation may have been caused by tectonism related to the bend at *ca* 1650 Ma on the apparent polar wander path. The timing and geochemistry of the alteration suggest that the fluids may have been related to those responsible for Pb–Zn mineralisation at Mt Isa.

The third component, C, has a dual polarity and moderate north-upward/south-downward inclinations (Figure 13f). C is pervasive throughout the formation and has a relatively narrow range of unblocking temperatures (typically 580–660°C) making it difficult to separate from components A (Figure 13e) and B. Dispersal of directions after bedding corrections indicates that C is post-deformational. Its mean direction, calculated after inverting the southerly directions, is declination 1.9°,

inclination –39.2°. This resembles the direction of overprint OP4.

Component D, giving the pole GPRR (Table 2), was found only in the ferricrete that marks the unconformity at locality L14 (Barr Hole). It unblocks between 660°C and 680°C and has a shallow west-upward inclination when tilt corrected (Figure 13g). As noted above, similar directions were found in the McArthur Basin correlative, the lower Masterton Sandstone, which therefore appears to have been hematised during the same depositional break.

PARADISE CREEK FORMATION

In contrast to the dominantly siliciclastic Gunpowder Creek Formation, the Paradise Creek Formation is mainly dolomitic with subordinate sandstone and shale. Its thickness at the type section along Paradise Creek is ~300 m (Southgate *et al.* 2000). U–Pb SHRIMP dates from zircons in redeposited tuffaceous layers range from 1658 Ma to 1653 Ma (Page *et al.* 2000), the older and more recently obtained ages being more consistent with the U–Pb zircon date of 1657 ± 3 Ma for the uppermost parts of the Paradise Creek Formation (Page *et al.* 2000).

Five widely separated localities were sampled; four of these were measured stratigraphic sections. They included the type section at Paradise Creek (locality L15), and the Esperanza Waterhole and Barr Hole sections near the Mt Gordon Fault where also the Gunpowder Creek Formation was sampled. The rock was predominantly grey. Remanence intensities were low (typically ~1 mA/m), and many samples either became magnetically unstable during thermal demagnetisation or yielded highly scattered to random directions. AF demagnetisation (some after thermal and liquid nitrogen pretreatments) failed to eliminate the scatter by selective removal of spurious remanence components. An exception was a group of nine grey dolomite samples from the upper part of the type section. The remanences of these samples unblocked at ~580°C, suggesting magnetite as carrier, and their directions formed a coherent group similar to those of overprint OP4 (Figure 14a). Although the mean remanence direction (declination 359.0°, inclination –39.6°) is based on few samples, the unblocking temperatures are high, indicating stable magnetisation, and each direction is well defined.

Another exception was a small component with south-southeast-downward inclinations (Figure 14b). This component was resolved in the temperature interval 250–400°C in 20 grey samples from two localities (L18 & L22, Figure 1). The relatively low unblocking temperature suggests that the remanence post-dates metamorphism in the Mt Isa Inlier, and a negative fold test confirms that the component is an overprint (Figure 14c, d). The grey colour of the samples suggests that the remagnetisation was not due to weathering. Its mean direction (declination 197.4°, inclination 55.8°) is consistent with the middle Mesoproterozoic apparent polar wander path, although that part of the path is very poorly defined. If so, this component may record an event during the assembly of Rodinia.

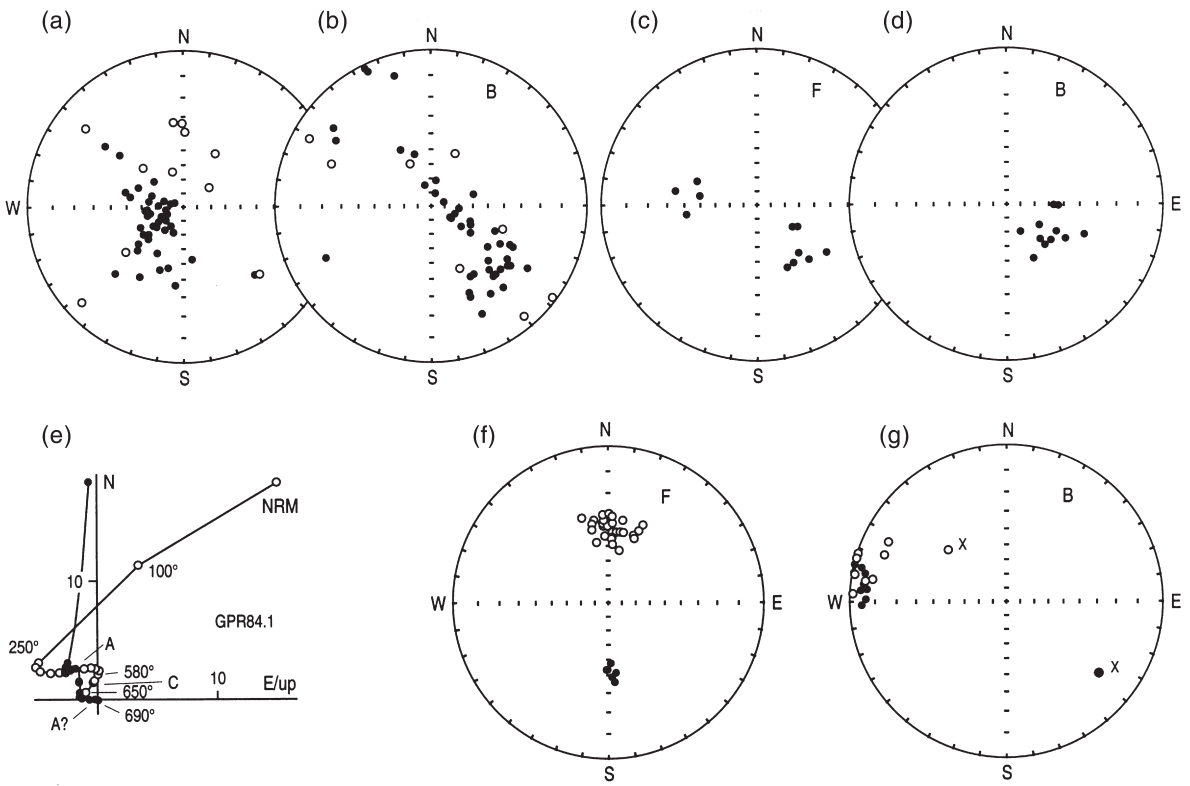


Figure 13 Gunpowder Creek Formation. Field (a) and bedding (b) corrected directions for component A. Dispersal of directions after bedding corrections indicates that A is an overprint. Field (c) and bedding (d) corrected directions for component B. Convergence of directions after bedding corrections indicates that B pre-dates deformation. Orthogonal projection plot (e) shows components A and C. Stereoplots (f) and (g) show the remanence directions for components C and D, respectively. Symbols as in Figure 3.

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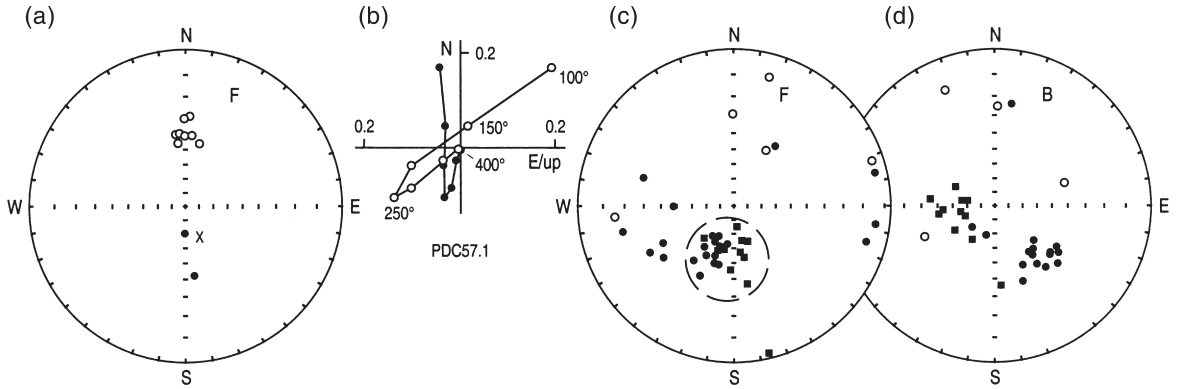


Figure 14 Paradise Creek Formation. Remanence directions shown in (a) are from Paradise Creek locality L20 (Figure 1). Orthogonal projection plot (b) shows a small south-southeast component, which unblocks at ~400°C and gives a negative fold test. Diagrams (c) and (d) show its remanence directions at localities L18 (circle symbols) and L22 (square symbols). The low unblocking temperature suggests that this component post-dates the Isan Orogeny. Outlier x in (a) was omitted from the pole calculation. Symbols as in Figure 3.

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ESPERANZA FORMATION

The stromatolitic Esperanza Formation comprises mainly siltstone and sandstone. Its thickness at the Gunpowder Creek type section is 250 m (Hutton & Wilson 1985). Ninety samples were collected from stratigraphic sections at two localities: Esperanza Waterhole (locality L23) and Mellish Park (locality L24). Most samples were grey, some were tan or red-brown. Both thermal and AF demagnetisation techniques were used, but all except the red-brown samples failed to give coherent directional groups. The latter samples were collected mainly from a zone of joint-related ferruginisation near the top of the Esperanza Waterhole section and close to the Mt Gordon Fault. The directions, shown in Figure 15, give a mean (declination 358.7°, inclination –42.7°) that corresponds to the overprint OP4.

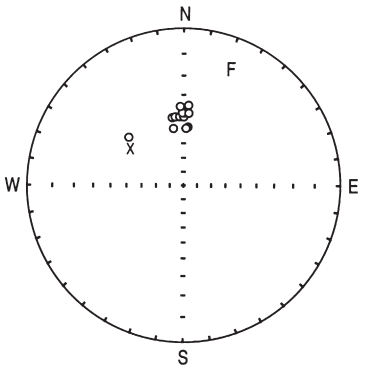


Figure 15 Esperanza Formation. Remanence directions from hematised samples. Direction marked x was excluded from the pole calculation. Symbols as in Figure 3.

LAWN HILL FORMATION

The Lawn Hill Formation at the top of the McNamara Group is divided into six siliciclastic members, Pmh1 (base) to Pmh6 (top). Its thickness at the type section (locality L25) is 1800 m. SHRIMP U–Pb dates from zircons in interbedded tuffs range from 1616 ± 5 Ma and 1611 ± 4 Ma for Pmh2 to 1591 ± 10 Ma for Pmh6 (Page *et al.* 2000). Exposures are generally poor.

Samples were collected at four localities from shales and siltstones ranging from dark grey to orange-brown and red. Demagnetisation was thermal, except for some pilot specimens that were demagnetised by AF (including thermal and liquid nitrogen pretreatments). AF demagnetisation failed to isolate any random components that may be responsible for directional scatter; thermal demagnetisation yielded coherent remanence directions for localities 25 (type section) and 26 (Silver King Pb–Zn mine area). Only the well-exposed Pmh6 was sampled at locality 25; Pmh1 and Pmh2 were sampled at locality 26.

Two remanence components, A and B, were resolved (Figure 16a). Component A, with steep southwest-downward inclinations (Figure 16a, b), unblocked between 440°C and 580°C and was found only at locality 26. Its directions cluster about a mean (declination 234.7°, inclination 66.9°) that resembles the direction of the overprint OP3. Component B was found at localities 25 and 26 (Figure 16a, c) where it occurred in grey as well as in cream and light red-brown samples. B unblocked between 500°C and 670°C and gave dual polarity directions. Two specimens contained both the normal and reversed components. Its mean direction (declination 5.1°, inclination –38.6°) is similar to that of OP4. A negative fold test implies that B is an overprint.

DISCUSSION

Apparent polar wander path

Figure 17 shows the updated apparent polar wander path. Its oldest part is based on only three poles: PCVB (Buddawadda Basalt), PCVI (dolerite intruding the basalt) and LNSO2 (Lochness Formation). Although their remanence directions are well grouped, each pole is based on a single site and on relatively few samples. This segment is therefore tentative as indicated in Figure 17 by the broken band. The Peters Creek Volcanics poles PCVB and PCVI are interpreted to be primary since they were determined from magnetite remanences and are well apart on the pole path (i.e. do not record a common regional overprint). Their ages are undetermined; however, the Buddawadda Basalt is

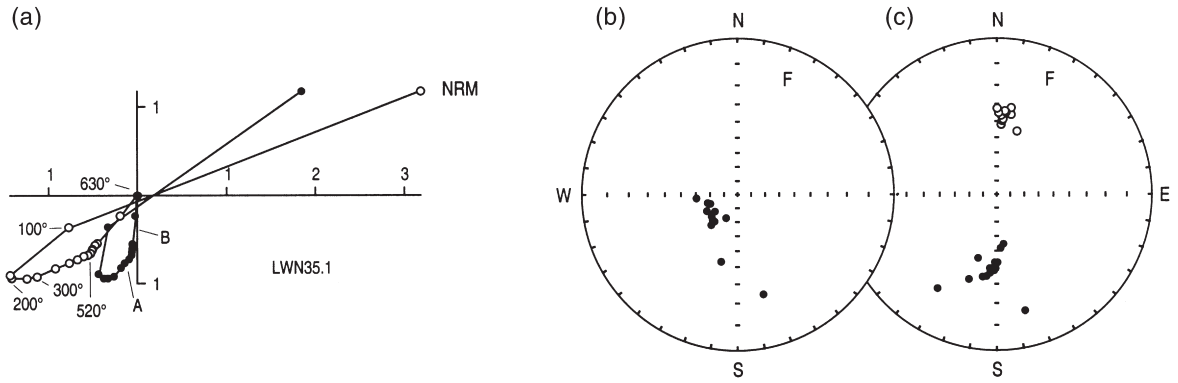


Figure 16 Lawn Hill Formation. Orthogonal projection plot (a) shows components A and B. Stereoplots (b) and (c) show the directions of the A and B components, respectively. Symbols as in Figure 3.

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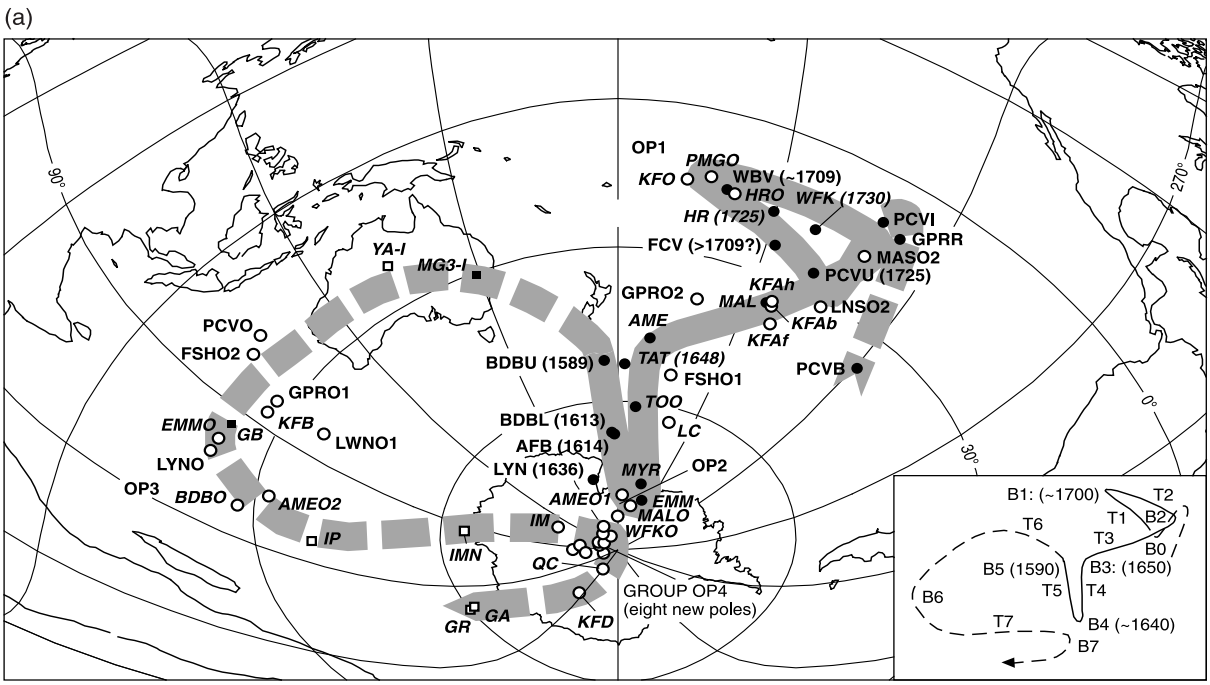
correlated with the Eastern Creek Volcanics of the Leichhardt Fold Belt (Scott *et al.* 1998) which have a maximum age of *ca* 1790 Ma given by the stratigraphically older Bottletree Formation (Page 1983). A minimum age of 1755 ± 3 Ma is provided by the Ballara Quartzite (Page 1998), which post-dates the Eastern Creek Volcanics (Jackson *et al.* 1990). The redbed pole LNSO2, lying in the 1730 Ma region of the apparent polar wander path, appears to be an overprint since the Lochness Formation is stratigraphically older than the Ballara Quartzite. LNSO2 may represent, for example, a weathering imprint imposed during a >25 million year hiatus at the sampling locality between the Lochness and Bigie Formations at the Calvert Superbasin boundary (Jackson *et al.* 2000).

Pole PCVU from the upper Peters Creek Volcanics marks the beginning of the better constrained apparent polar wander path. PCVU, WFK (lower Wollogorang Formation: Idnurm *et al.* 1995) and WBV (West Branch Volcanics) have well-defined positions, ages and tectonic settings, and are therefore key poles for the period 1730–1710 Ma. This path segment, starting at LNSO2, covers the initial phase of deposition in the Calvert Superbasin. The segment ends at the group of poles HRO, PMGO, and KFO that record overprinting due to the passage of fluids at the southeastern and northwestern margins of the McArthur Basin (Giddings & Idnurm 1993*;* Idnurm *et al.* 1995). Poles HRO and PMGO were previously attributed to regional K-metasomatism (Idnurm *et al.* 1995), but it was not clear if the fluids were mobilised by late-stage igneous activity, by post-magmatic tectonism (Solomon & Heinrich 1992) or some other event. The new pole from the *ca* 1709 Ma West Branch Volcanics and the revised 1725 ± 2 Ma age for the Hobblechain Rhyolite now jointly rule out the metasomatism as a late-stage magmatic process. This therefore leaves the possibility that the fluids were driven by tectonism, which may be related to the bend in the apparent polar wander path (see below). The same tectonic event may have initiated deposition of the Prize Supersequence in the Calvert Superbasin (Southgate *et al.* 2000).

The next two poles, MASO2 (lower Masterton Sandstone overprint) and GPRR (ferricrete developed on top of the lower Gunpowder Creek Formation), indicate that the region had returned to approximately its pre-1730 Ma latitude and moreover was oriented so that it faced the geographic pole the same way as previously. Neither pole is dated. However, the location of GPRR on the apparent polar wander path and the stratigraphic position of the ferricrete in the Gun 1 sequence (Southgate *et al.* 2000) indicate that the ferricrete formed towards the end of the hiatus within the Gunpowder Creek Formation. This makes the ferricrete broadly equivalent in age to the Moondarra Siltstone (1668 ± 8: Page *et al.* 2000), but probably younger than the Carters Bore Rhyolite (1678 ± 2 Ma) and Sybella Granite (1671 ± 8 Ma); three formations to the east of the Mt Gordon Fault (Southgate *et al.* 2000). The age of the pole GPRR is therefore likely to be *ca* 1670 Ma.

The apparent polar wander path now takes an east-southeast course towards the group of primary poles AME (Amelia Dolomite), TAT (Tatoola Sandstone) and TOO (Tooganinie Formation) that cluster around bend 3 on the path. Before reaching the bend the path passes through the previously determined primary pole from the lower Mallapunyah Formation (MAL) and the cluster of overprint poles KFA from the Kombolgie Formation. A recent date of 1648 ± 3 Ma for the Tatoola Sandstone (Page *et al.* 2000) provides the only age constraint for this part of the apparent polar wander path. Two new overprint poles FSHO1 (Fish River Formation near the Fish River Fault) and GPRO2 (Gunpowder Creek Formation close to the Mt Gordon Fault) plot near the *ca* 1650 Ma pole cluster. The timing is similar to that of the Mt Isa Pb–Zn deposit (Sun *et al.* 1994; Carr *et al.* 1996), suggesting that instead of being restricted to the Mt Isa Fault region, fluid circulation along faults in the Lawn Hill Platform was much more widespread.

AME–TAT–TOO mark a southward turn of the apparent polar wander path towards the primary poles MYR (Myrtle Shale) and EMM (Emmerugga Formation), and the cluster of overprint poles WFKO, MALO and AMEO1. These five poles form the apex of a hairpin bend, B4, which appears to coincide with the *ca* 1640 Ma period of fluid flow that created the HYC Pb–Zn deposit in the McArthur Basin. Passing the bend, the path takes a north-northeast route defined by the new and revised poles LYN (Lynott Formation, 1636 ± 4 Ma), AFB (Amos Formation, 1614 ± 4 Ma), BDBL (lower Balbirini Dolomite, 1609 ± 3 Ma and 1613 ± 4 Ma) and BDBU (upper Balbirini Dolomite, 1589 ± 3 Ma). This palaeomagnetically and geochronologically best-defined segment of the north Australian Proterozoic apparent polar wander path continues to



POLE ACRONYMS

AFB - Amos Formation
AME, AMEO1 (OP2), AMEO2 (OP3) - Amelia Dolomite
BDBL, BDBU, BDBO (OP3) - Balbirini Dolomite
EMM, EMMO (OP3) - Emmerugga Dolomite
ESPO (OP4) - Esperanza Formation
FCV, FCVO (OP4) - Fiery Creek Volcanics
FSHO1, FSHO2 (OP3A), FSHO3 (OP4) - Fish River Formation
GA, GB - Gawler Craton dykes
GR - Gawler Volcanics
GPRO1 (OP3A), GPRO2, GPRO3 (OP4), GPRR - Gunpowder Creek Formation
HR, HRO (OP1) - Hobblechain Rhyolite
IM - Mount Isa Inlier dykes (overprint)
IMN - Iron Monarch ore
IP - Iron Prince ore
MG3-I - Mt Goldsworthy ore
KFAb, KFAf, KFAh, KFB, KFD, KFO (OP1) - Kombolgie Formation overprints
LC - Lunch Creek Gabbro
LNSO1 (OP4), LNSO2 - Lochness Formation
LWNO1 (OP3A), LWNO2 (OP4) - Lawn Hill Formation
LYN, LYNO (OP3) - Lynott Formation
MAL, MALO (OP2) - Mallapunyah Formation
MASO (OP4), MASO2 - Masterton Sandstone
MLSO - Mount Les Siltstone
MYR - Myrtle Shale
PCVU, PCVO (OP3A), PCVB, PCVI - Peters Creek Volcanics
PDCO1 (OP4), PDCO2 - Paradise Creek Formation
PMGO (OP1) - Packsaddle Microgranite
QC (OP4) - Quamby Conglomerate
TAT - Tatoola Sandstone
TOO - Tooganinine Formation
WBV - West Branch Volcanics
WFK, WFKO (OP2) - Wollogorang Formation
YA-I - Yilgarn Block dykes

POLES FROM NORTHERN AUSTRALIAN CRATON
● Primary
○ Overprint

POLES FROM OTHER AUSTRALIAN CRATONS
■ Primary
□ Overprint

Figure 17 Diagrams (a) and (b) show the Late Palaeoproterozoic – Early Mesoproterozoic apparent polar wander path for northern Australia. Acronyms in italics indicate poles obtained previously, acronyms in normal font indicate new or updated poles. Numbers in brackets after acronyms give U–Pb zircon ages in Ma. Ellipses around the poles in (b) are 95% confidence limits. Parts of the apparent polar wander path regarded as tentative are shown by a broken band. Inset shows interpretation of apparent polar wander path in terms of tracks (T) and bends (B) as discussed in the text.

the beginning of the Isan Orogeny at *ca* 1600 Ma when deposition ceased in northern Australia. The only palaeomagnetic records for the ~100 million year orogeny are undated magnetic overprints.

After BDBU the path leads west across Australia towards an arcuate band of poles in the Indian Ocean. The trans-Australia segment is constrained by poles MG3-I (Mt Goldsworthy iron ore) and YA-I (Yilgarn dykes). Both poles have very poor age constraints. Moreover, the use of poles from other cratons to fill the gap requires the untested assumption that the Australian craton had been assembled by the Early Mesoproterozoic. Consequently, the location of the trans-Australia segment is tentative. The poles to the west of Australia represent overprints in the McArthur Basin and Lawn Hill Platform. Their spread appears to be due to a time lag in overprinting between the McArthur

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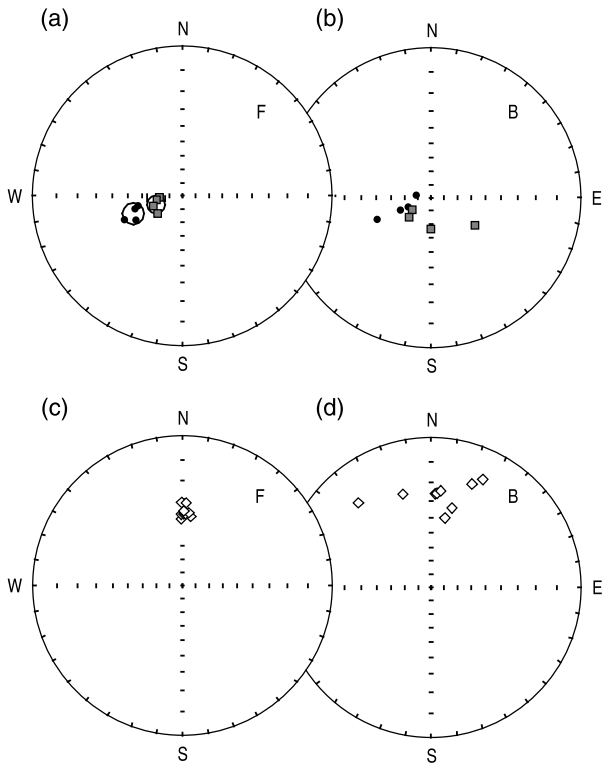


Figure 18 Field (a) and bedding (b) corrected formation mean directions for overprints OP3 (circle symbols) and OP3A (square symbols). Also shown in (a) are 95% confidence circles for the overall mean directions of OP3 and OP3A. Plots (c) and (d) show, respectively, the field and bedding corrected directions for overprint OP4. Large increase in directional scatter after bedding corrections suggests that these overprints post-date the main deformational phase of the Isan Orogeny. Symbols as in Figure 3.

Basin and Lawn Hill Platform, the Lawn Hill Platform poles lying on the older part of the apparent polar wander path segment. It should be noted that the 95% confidence circles of the respective mean remanence directions do not intersect (Figure 18a), confirming that the difference between the pole groups is statistically significant. The groups therefore seem to record two separate events, OP3A (Lawn Hill Platform) and OP3 (McArthur Basin). Comparison of the remanence directions before (Figure 18a) and after (Figure 18b) bedding corrections indicates that OP3 and OP3A were acquired after the main phase of deformation (D_3) in the Mt Isa Inlier. U–Pb ages of zircons from two generations of pegmatite provide 1532 ± 7 Ma and 1480 ± 14 Ma as the maximum and minimum age limits for D_3 (Connors & Page 1995). A possible explanation for the OP3A/OP3 age difference is that hematite-bearing fluids were mobilised by stresses in the late stages of the Isan Orogeny, and that these propagated from the east to the west.

After the pole group OP3, the path traverses a sparsely populated stretch containing only undated poles from the Iron Prince and Iron Monarch ore deposits, both in South Australia. The next pole, IM, was obtained from metamorphosed mafic dykes in different parts of the Mt Isa Inlier, and therefore represents a regional event. Because the dykes are not hematised, IM was assumed to represent thermal remanence acquired by magnetite during the last phase of metamorphism at the Mt Isa Inlier, at 1550–1500 Ma or later, depending on the rate of regional cooling (Tanaka & Idnurm 1994).

Tightness of the next overprint pole group, OP4, before bedding corrections (Figure 18c) in contrast with scatter after the corrections (Figure 18d) demonstrates that the overprints post-date the Isan Orogeny. OP4 was found at various sampling localities across the Lawn Hill Platform,

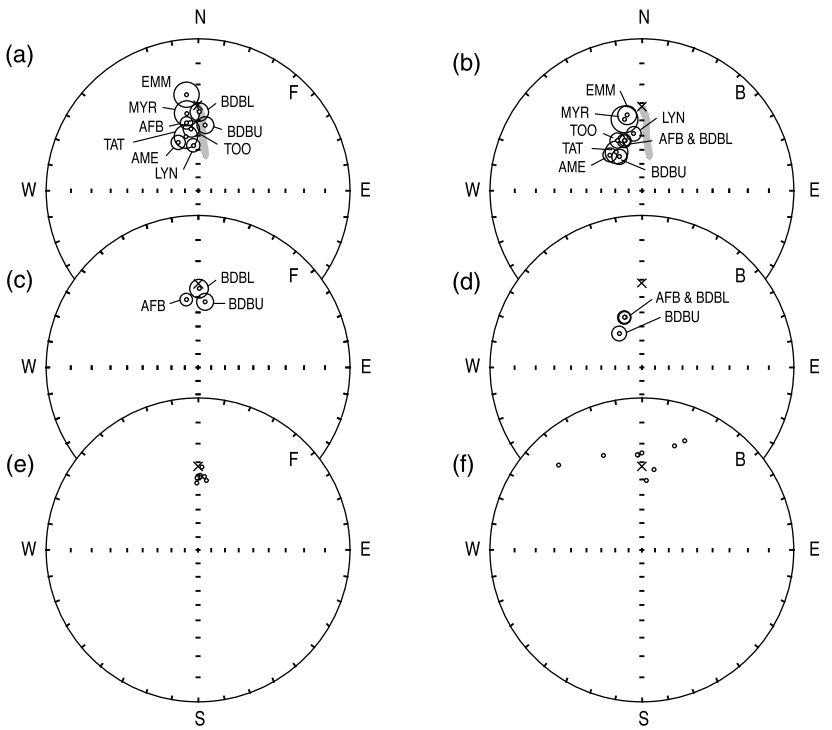


Figure 19 Stereodiagrams (a) and (b) compare the mean remanence directions and their 95% confidence circles of the 1650–1600 Ma McArthur Basin formations with the trajectory of Cenozoic dipole directions in the southern McArthur Basin (grey stipple). The Cenozoic trajectory is calculated from the apparent polar wander path of Idnurm (1994). Note that with the exception of BDBL and BDBU the directions lie west of the Cenozoic trajectory. Plots (c) and (d) compare the mean remanence directions of the Amos Formation with those of the upper and lower Balbirini Dolomite. Before bedding corrections (c) the AFB and BDBL directions differ; after corrections (d) they coincide. Plots (e) and (f) show the mean overprint directions of formations that contribute to the overprint group OP4. AFB, Amos Formation; AME, Amelia Dolomite; BDBL, lower Balbirini Dolomite; BDBU, upper Balbirini Dolomite; EMM, Emmerugga Dolomite; LYN, Lynott Formation; MYR, Myrtle Shale; TAT, Tatoola Sandstone; TOO, Tooganinie Formation. F and B indicate field-corrected (*in situ*) and bedding-corrected directions, respectively. Cross indicates present dipole field direction.

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but has been identified in the McArthur Basin in only the ferruginised uppermost Masterton Sandstone. Elsewhere too, OP4 is mainly associated with hematite, which together with its proximity to the South Pole suggests recent weathering as the cause. Against this possibility: (i) not all OP4 poles were obtained from distinctly hematised outcrops (e.g. Mt Les Siltstone); (ii) OP4 was found in widely different geomorphic settings from creek beds to cliff faces; and (iii) such a recent age of weathering is unusual in other parts of Australia where palaeomagnetic records indicate Pliocene or earlier weathering peaks (Schmidt & Embleton 1976; Idnurm & Senior 1978). OP4 is therefore interpreted as Mesoproterozoic, but post-dating the gently folded Roper and South Nicholson Groups.

An overprint pole from the Kombolgie Formation and two overprint poles from the Gawler Craton extend the apparent polar wander path northward, indicating a prominent bend at OP4. The extension is, however, tentative since these overprints do not have minimum age constraints and could belong to a much younger part of the path. The bend and overprint OP4 suggest a continuation of Mesoproterozoic tectonism and fluid movement in northern Australia after the Isan Orogeny.

Question of Cenozoic overprinting

Assignment of palaeomagnetic poles to their respective apparent polar wander path segments can be problematic if the poles plot where different parts of the apparent polar wander path either intersect or approach each other. To determine which path segment a pole belongs to requires age constraints on its remanence. The ambiguity in pole assignment arises from recurrence of the same or a similar geomagnetic dipole field direction at different times in the geologic history of the sampling region.

Bends 4 and 7 of the present apparent polar wander path occur close to the South Pole. Likewise, the remanence directions that define tracks 4 and 5 resemble broadly the Cenozoic dipole directions (Figure 19a). In view of the prevalence of Cenozoic chemical weathering imprints in Australia, could these poles represent Cenozoic overprints rather than Proterozoic magnetisations?

Several lines of evidence suggest otherwise. First, although the northern Australian remanence directions resemble those of Cenozoic fields, all except the directions from the Balbirini Dolomite are displaced to the west of the Cenozoic directions (Figure 19a). The displacement is substantial, making a Cenozoic origin of the remanences unlikely. The two exceptions are discussed below.

Second, the Mallapunyah and Tooganinie Formations as well as the Emmerugga Dolomite yield positive fold tests (Idnurm *et al.* 1995). Therefore, these three remanences, which determine the general shape of track 4, pre-date regional deformation in the southern McArthur Basin. The intensity of the Isan Orogeny in the adjacent Mt Isa Inlier suggests that the deformation is mainly Isan (*ca* 1600–1500 Ma).

Third, comparison of the Balbirini Dolomite and Amos Formation remanence directions suggests that at least two of these remanences are not Cenozoic (Figure 19c, d). As discussed above, Watson's test (1956) indicates that the mean directions of the upper and lower Balbirini Dolomite differ at >99% significance level. Likewise, the field corrected mean direction from the Amos Formation differs significantly from those of both the upper and lower Balbirini Dolomite because its 95% confidence circle does not intersect the confidence circles for the dolomite. Because the sampling sites for these three formations were located in similar terrains and within ~1 km of each other, diachronism of Cenozoic weathering seems an unlikely explanation for the directional differences. Moreover, diachronism would not explain why the remanence directions of the similarly aged Amos Formation and lower Balbirini Dolomite differ before bedding corrections but agree after the corrections (Figure 19d). Although these comparisons do not support the Cenozoic origin of the Amos Formation and lower Balbirini Dolomite poles, they leave open the age of the upper Balbirini Dolomite pole. However, as noted earlier, BDBU pole fails only marginally to return a positive fold test which would confirm its pre-deformational origin. In addition, the upper Balbirini pole continues the northward course of the apparent polar wander path, consistent with a primary origin.

Watson's test (1956) for the *in situ* mean remanence directions of the other formations that define tracks 4 and 5—AME, TAT, MYR and LYN in Figure 19a—indicates a $<<1\%$ probability that these directions come from the same population, i.e. represent a common, regional Cenozoic overprint. (Although the test is not rigorous because the precision parameters, k, of the directions differ, the low level of significance suggests that even if a rigorous test were available the result is likely to be negative.) Alternatively, could the poles be Cenozoic with directional spread attributed to diachronism of weathering? That possibility seems unlikely because three of the poles come from similar terrains located within a small area. It should also be noted that these four poles plot in stratigraphic sequence with the other poles (Idnurm *et al.* 1995), consistent with primary magnetisation.

Concerning the poles WFKO, MALO and AMEO1 that form the overprint group OP2, we note first that mottling and other macroscopic indicators of weathering were absent in the samples. Second, most of the AME samples were silicified and therefore impermeable (Idnurm *et al.* 1995). The samples for these poles were collected from a variety of lithologies (stromatolitic dololutites to fine-grained sandstones) and landscape settings (riverbeds to cliff faces). Their only common denominator was that they were obtained exclusively from locations adjacent to prominent north to north-northwest faults. The lack of mottling, impervious nature of some samples and association of OP2 with fault zones is suggestive of ancient fluid movement along faults rather than Cenozoic weathering.

As noted above, the overprint group OP4 lies close to the South Pole, consistent with latest Cenozoic, possibly Late Pliocene, weathering. However, other evidence is less supportive of weathering. OP4 has been found within a large region, in a wide range of landscape settings, and in different lithologies. It is common in the Lawn Hill Platform but is rare in similar terrains in the McArthur Basin. The sampling site of the Masterton Sandstone, where OP4 was found in the McArthur Basin, contains ferruginous mottles whose distribution is controlled by facies and bedding planes suggesting that they originated from post-lithification groundwater movement (Jackson *et al.* 1987).

Some of the Masterton Sandstone mottles are enclosed by silica (Jackson *et al.* 1987 figure 40). Many samples show no traces of hematite pigment. The mean remanence directions of OP4 in the different formations form a close cluster (Figure 19e, f), indicating the absence of diachronism despite a wide range of geomorphic settings from riverbeds to cliff faces, and the large sampling region. Perhaps most significantly, a positive fold test for an overprint in the Quamby Conglomerate in the eastern Mt Isa Inlier indicates that its remanence is pre-Cenozoic (Idnurm & Wyborn 1998). The Quamby Conglomerate was sampled in an open-cut mine in hilly terrain and its overprint direction (declination 3.9°, inclination –31.2°, $\alpha_{95} = 5.5°$) resembles OP4. For these reasons OP4 is interpreted as Proterozoic.

While the evidence does not favour a Cenozoic age for the poles, contributions to the remanences from Cenozoic weathering cannot be ruled out because the similarity of directions makes any contribution very difficult to detect. This applies especially to the Balbirini Dolomite where the *in situ* remanence directions are indistinguishable from Cenozoic directions. The large scatter of the Tooganinie Formation directions could likewise be due to unresolved Cenozoic components of reverse polarity.

Geodynamic implications

As discussed by Loutit *et al.* (1994), the Late Palaeoproterozoic apparent polar wander path for Australia contains prominent inflections or bends that represent changes in the direction of plate movement and imply reorganisation of stresses within the plate. A change in the stress field can have a major impact on basin architecture through flexure and faulting. It can also play a vital role in forming ore deposits by mobilising mineralised fluids within basins (Loutit *et al.* 1994).

The bends join quasilinear segments, or tracks, which can be approximated on the globe by arcs of small circles or, more accurately, by a sequence of small, near-concentric arc elements. Polar wander along such a track therefore corresponds to a rotation about the axis that passes through the centres of the earth and the small circle. In the normal geographic reference frame where it is the plate rather than the geographic pole that moves, the movement is a rotation about the same axis but in opposite sense to the apparent polar wander.

Five tracks (T) and five bends (B) are identified on the apparent polar wander path between the Lochness Formation overprint pole LNSO2 and the upper Balbirini Dolomite pole BDBU (Figure 17), and at least two tracks and three bends are likely between BDBU and the pole group OP4. (The two poles that pre-date LNSO2 are too few to reliably define what may be a complex older part of the apparent polar wander path.) As discussed by Southgate *et al.* (2000), most of the apparent polar wander path bends appear to be associated with supersequence boundaries in the Calvert and Isa Superbasins. Similarly, Wyborn *et al* (1998a) have proposed a causal link between the bends and the emplacement of major batholiths in the Late Palaeoproterozoic – Early Mesoproterozoic: the Weberra Granite being associated with bend 1, Sybella suite with bend 2, Mt Webb Granite (Western Australia) with bend 4, and the Hiltaba suite (South Australia) with bend 5.

Track 1 covers the initial stages of formation of the Calvert Superbasin, which were accompanied by widespread bimodal volcanism and granite intrusion. The granites and predominantly felsic volcanics crop out along a broad belt, starting in the southeast with the 1741 Ma Gin Creek Granite and ending in the northwest with the *ca* 1709 Ma West Branch Volcanics (GCG and WBV, Figure 20). U–Pb zircon dates suggest a decreasing age trend towards northwest. Is this trend real or purely coincidental, reinforced by a sparse distribution of Proterozoic outcrop which is, moreover, itself restricted to a northwest–southeast belt? If real, the age decrease may be explained by a long-lived thermal anomaly in the upper mantle that left a trail of volcanics and granite intrusions as the plate moved across it along a southeast course. Palaeomagnetic evidence supports that possibility. First, the northwest age decrease of igneous activity is consistent with the northwest direction of apparent polar wander along track 1. Indeed, the agreement between the igneous outcrops and polar wander may extend beyond track 1 to the 1748 Ma

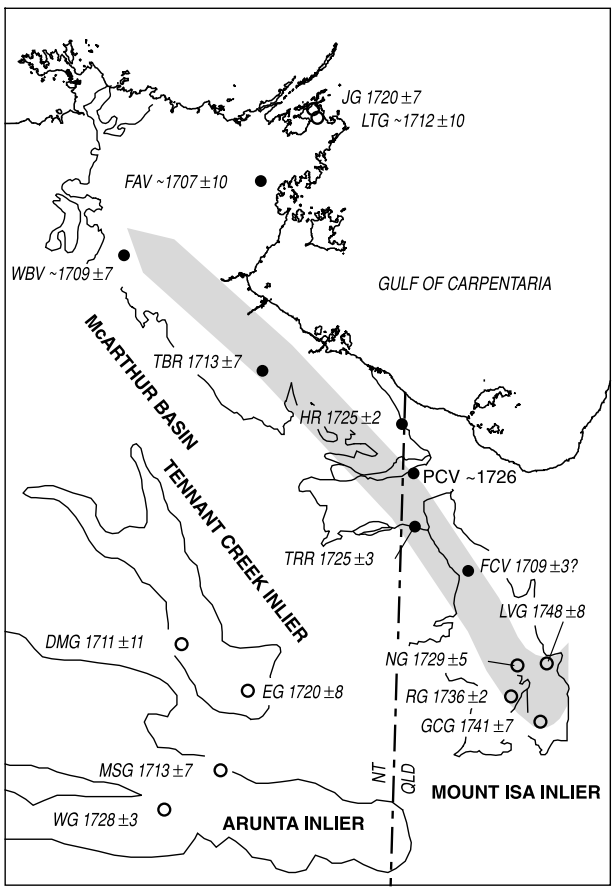


Figure 20 Distribution of 1750–1700 Ma volcanics (filled symbols) and granites (open symbols) from Scott *et al.* (2000). Arrow shows decreasing age trend of igneous units in the McArthur Basin and Mt Isa Inlier. DMG, Devils Marbles Granite; EG, Elkedra Granite; FAV, Fagan Volcanics; FCV, Fiery Creek Volcanics; GCG, Gin Creek Granite; HR, Hobblechain Rhyolite; JG, Jimbu Granite; LTG, Latram Granite; LVG, Levian Granite; MSG, Mt Swan Granite; NG, Natalie Granite; PCV, Peters Creek Volcanics; RG, Revenue Granite; TBR, Tanumbirini Rhyolite; TRR, Top Rocky Rhyolite; WBV, West Branch Volcanics; WG, Wuluma Granite.

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Levian Granite (LVG) on the north–south-oriented apparent polar wander path segment that precedes bend 0. Second, the poles of rotation for track 1 and the igneous band are similar, and the angular lengths of the respective arcs agree within the confidence limits of the palaeomagnetic poles (Figure 21). The agreement would be improved further if the age of the Fiery Creek Volcanics at the palaeomagnetism sampling site were to turn out ≥1725 Ma (Scott *et al.* 1998; also discussion above). However, it should be noted that granites of similar ages in the Tennant Creek and Arunta Inliers to the south do not show such an age trend (Figure 20). This may reflect a much shorter life span of their associated thermal anomalies, or some different mechanism of emplacement.

The same, or similar, rotation axes as for track 1 also describe the movement of the plate for tracks 4 and 5, and possibly 2 (Figure 21). Therefore, the plate movement during the formative ~150 million years period of the Calvert and Isa Superbasins may be described in terms of clockwise–anticlockwise rotations about Euler rotation poles located southwest of Australia. An exception is the period *ca* 1670–1650 Ma when the rotation pole had switched to a location northeast of Australia. In an ocean–continent setting such behaviour would correspond to the Wilson cyclicity of repeated ocean opening and closure (Wilson 1966). Its cause is not clear, but this style of movement is difficult to understand in terms of plate boundary forces such as slab pull or ridge push. A simple explanation may involve alternate coupling of the lithosphere to oppositely convecting cells in the mantle. The loss or transfer of traction may perhaps have been related to the thermal state of the upper mantle. Scott *et al.* (2000) conclude from the magnitude and extent of igneous activity in northern Australia that the temperature of the mantle below the region was high for ~200 million years.

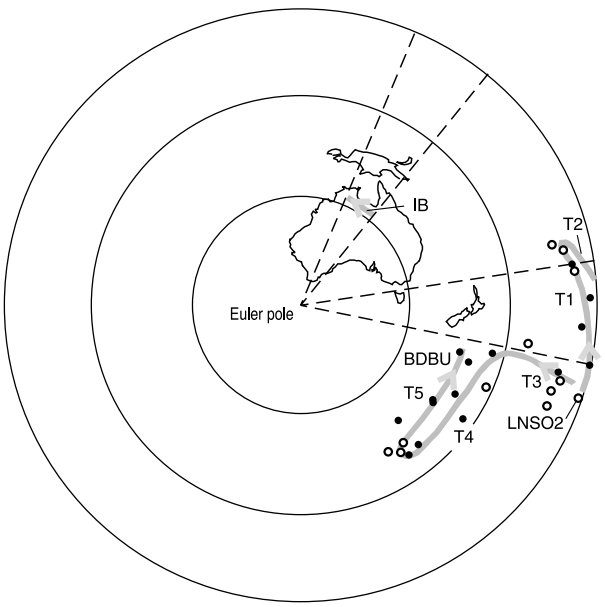


Figure 21 Australian apparent polar wander path for the period 1740–1710 Ma shown on an equal-area stereographic projection. The projection is centred on an Euler (rotation) pole that describes approximately the rotation of central northern Australia as represented by tracks 1, 4, 5 and possibly 2. IB is a band of igneous bodies with a decreasing age trend towards the northwest. Broken lines indicate angles subtended at the Euler pole by IB and the 1740–1710 Ma apparent polar wander path segment. As discussed in the text, the concentricity of IB and the apparent polar wander path segment, the similarity of their subtended angles, and the agreement between the direction of age decrease in IB and that of apparent polar wander, suggest that the 1740–1710 Ma igneous activity may have been caused by the movement of northern Australia across an upper mantle thermal anomaly.

As discussed above and more fully by Southgate *et al.* (2000), bends 2–4 are associated with Pb–Zn mineralisation at Broken Hill, Mt Isa and HYC, respectively. The existence of bend 5 is inferred from the requirement that the track has to pass through the pole groups OP3A and OP3 in the Indian Ocean, and that this would need a change in course of the apparent polar wander. The location of bend 5 is uncertain because of the lack of palaeomagnetic data for the period 1590–1560 Ma. In Figure 17 bend 5 is tentatively drawn close to the 1589 Ma pole BDBU on the assumption that the bend marks the onset of the Isan Orogeny. (Since track 5 remains linear up to BDBU, the orogeny appears to post-date 1589 ± 3 Ma.) Most other bends on the late Proterozoic apparent polar wander path appear to be associated with fluid circulation, suggesting that bend 5 may likewise have been accompanied by fluid movement. This is broadly consistent with the Pb isotopic model age of *ca* 1575 Ma for mineralisation at the Century Zn deposit (Carr *et al.* 1996). Alternatively, bend 5 may be related to the D_1 phase of Isan Orogeny and the setting up of the subbasin that hosts the Century deposit. The Century deposit itself may therefore correspond to a later, as yet unrecognised, bend northwest of bend 5, and the commencement of the D_2 main phase of deformation in the Mt Isa Inlier.

CONCLUSIONS

Ten new primary poles reinforce and extend the Late Palaeoproterozoic – earliest Mesoproterozoic apparent polar wander path for northern Australia. Concurrently, a detailed time calibration has become available for the apparent polar wander path from new and recently published U–Pb zircon SHRIMP dates for nine of the formations that have yielded primary poles. These dates help to develop the apparent polar wander path into a powerful tool for the study of one of the most important periods of Proterozoic mineralisation in Australia. Applications include dating, correlation and tracing of the pathways of mineralising fluids.

The upgraded Late Palaeoproterozoic apparent polar wander path comprises five quasilinear segments or tracks, and five bends. The shapes and orientations of the tracks suggest that the movement of the McArthur Basin – Lawn Hill Platform region may be described as a succession of clockwise–counterclockwise rotations about mostly similarly located Euler poles southwest of Australia. Palaeomagnetic evidence suggests that during the period 1740–1710 Ma the region passed across a long-lived thermal anomaly in the upper mantle.

Each of the Late Palaeoproterozoic apparent polar wander path bends may be related to a supersequence

boundary in northern Australia. Most bends appear to be associated with magnetic overprints due to fluid flow, three of these being associated with the formation of world-class Pb–Zn orebodies. Thus palaeomagnetic evidence suggests that global tectonic events, rather than local tectonic readjustments or eustatic changes in sea-level, governed much of the basin development and fluid mobilisation in the Late Palaeoproterozoic of northern Australia.

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